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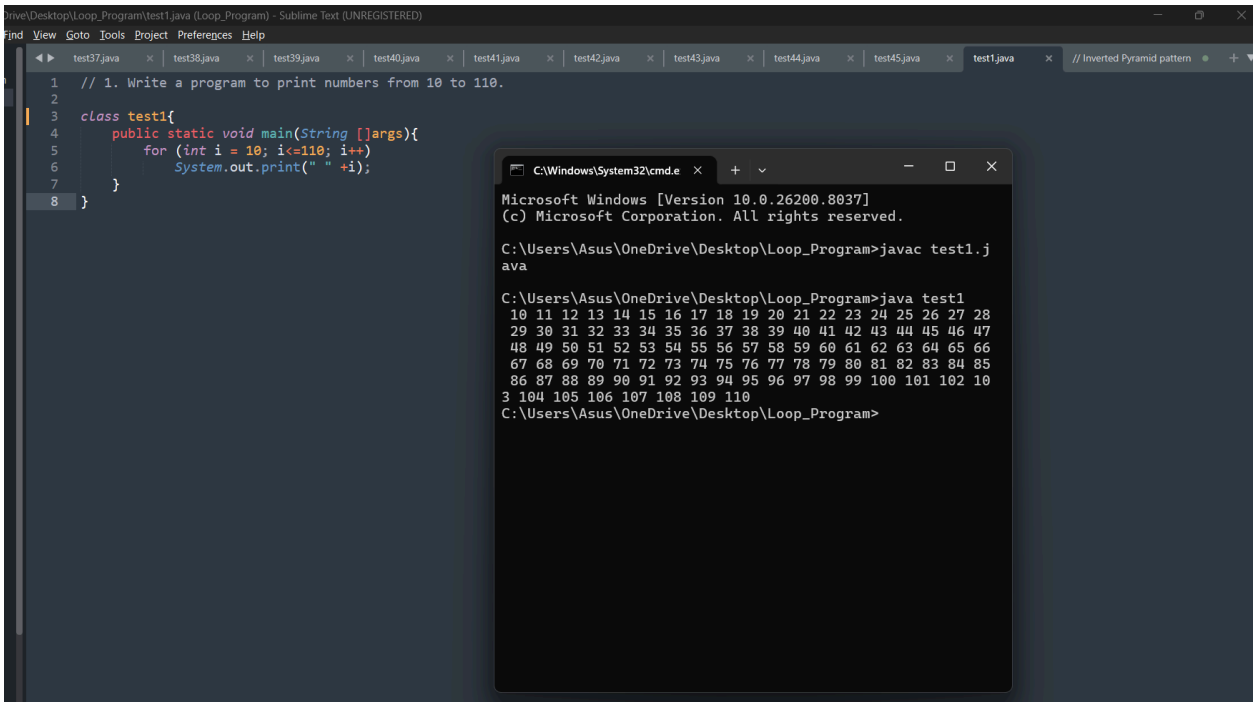
Batch: PG-DAC February 2026
Subject: JAVA
Assignment: 3
Form Number: 251101107

// 1. Write a program to print numbers from 10 to 110.
Solution:

```
class test{
        public static void main(String []args){
                for (int i = 10; i<=110; i++)
                        System.out.print(" " +i);
        }
}
```



// 2. Write a program to calculate the sum of all numbers from 1 to 100.
Solution:

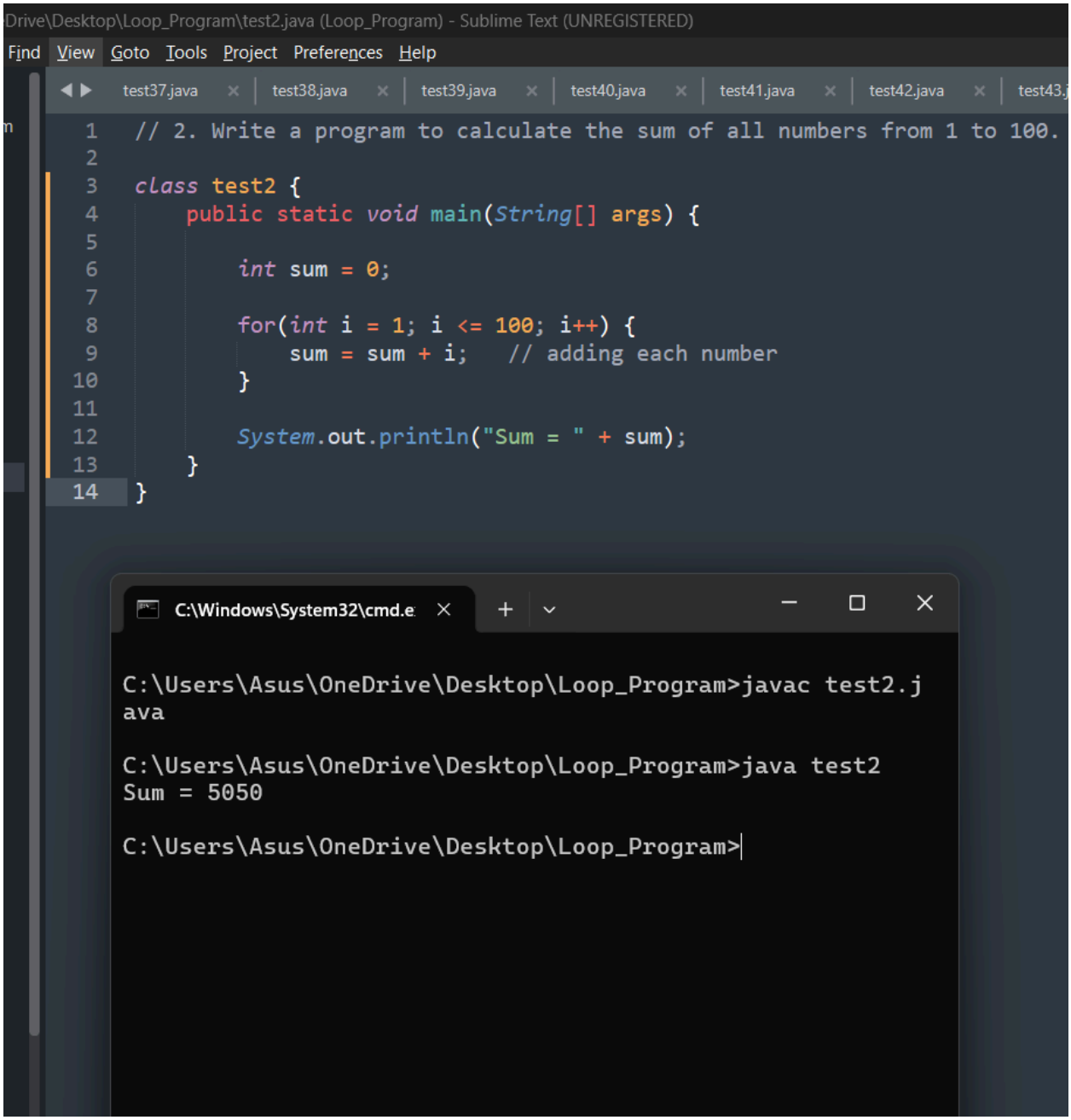
```
class test2 {
   public static void main(String[] args) {

      int sum = 0;

      for(int i = 1; i <= 100; i++) {
         sum = sum + i;   // adding each number
```

```
        }

        System.out.println("Sum = " + sum);
    }
}
```



// 3. Write a program to print the multiplication table of a given number.
Solution:

```
class test3{
	public static void main(String []args){
		int n = Integer.parseInt(args[0]);
		for (int i = 1; i<=10; i++)
			System.out.println(n + " * "  + i +" = "+(n * i));
	}
}
```

```java
// 3. Write a program to print the multiplication table of a given number.

class test3{
    public static void main(String []args){
        int n = Integer.parseInt(args[0]);
        for (int i = 1; i<=10; i++)
            System.out.println(n + " * "  + i +" = "+(n * i));
    }
}
```

```
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C:\Users\Asus\OneDrive\Desktop\Loop_Program>javac test3.j
ava

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test3 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

C:\Users\Asus\OneDrive\Desktop\Loop_Program>
```
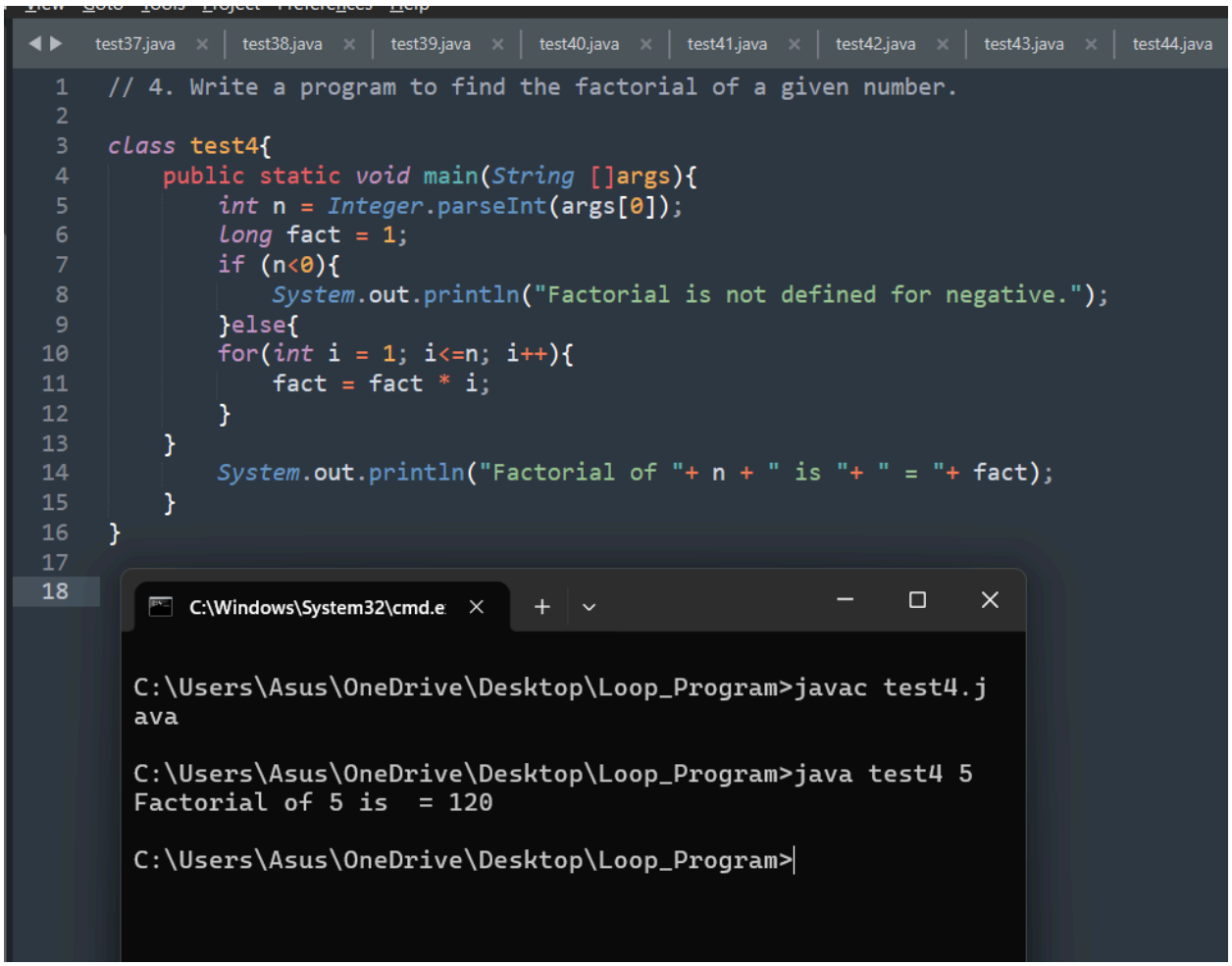

// 4. Write a program to find the factorial of a given number.
Solution:

```java
class test4{
        public static void main(String []args){
                int n = Integer.parseInt(args[0]);
                long fact = 1;
```

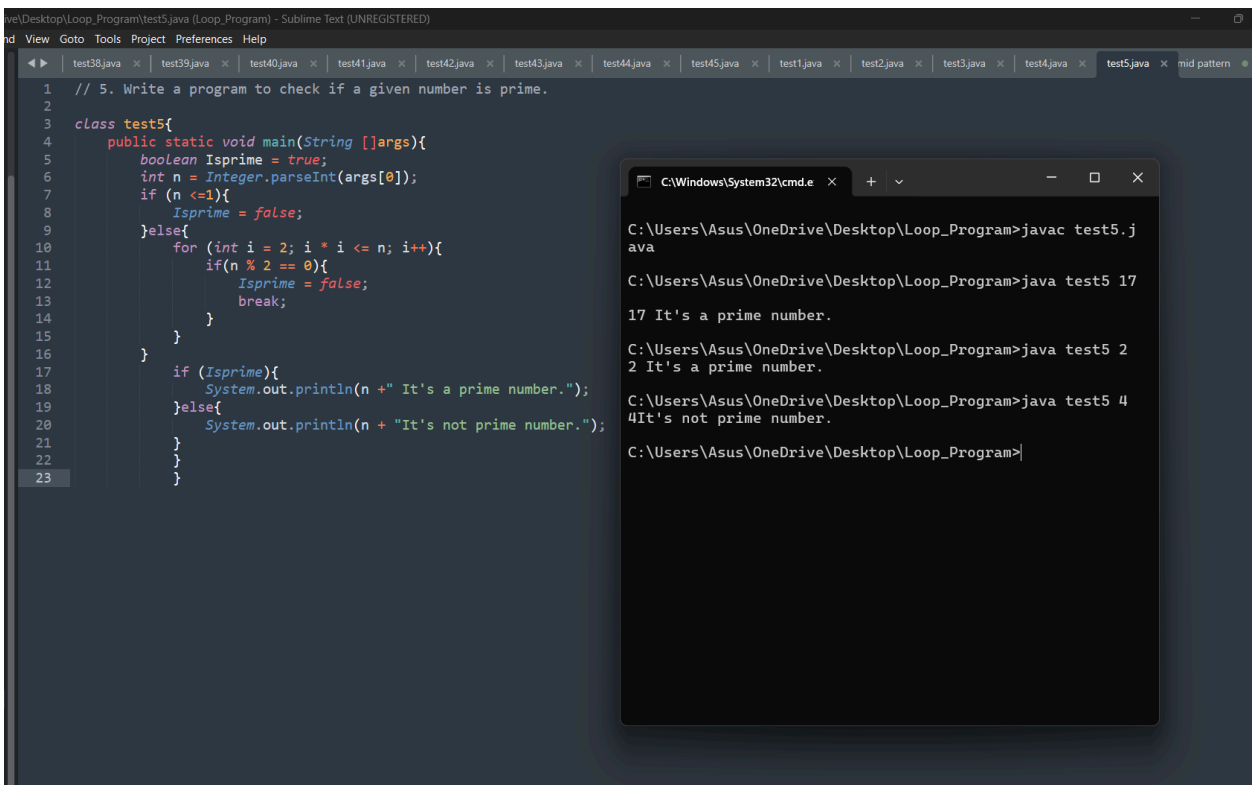
```
            if (n<0){
                System.out.println("Factorial is not defined for negative.");
            }else{
            for(int i = 1; i<=n; i++){
                fact = fact * i;
            }
        }
            System.out.println("Factorial of "+ n + " is "+ " = "+ fact);
        }
}
```



// 5. Write a program to check if a given number is prime.
Solution:

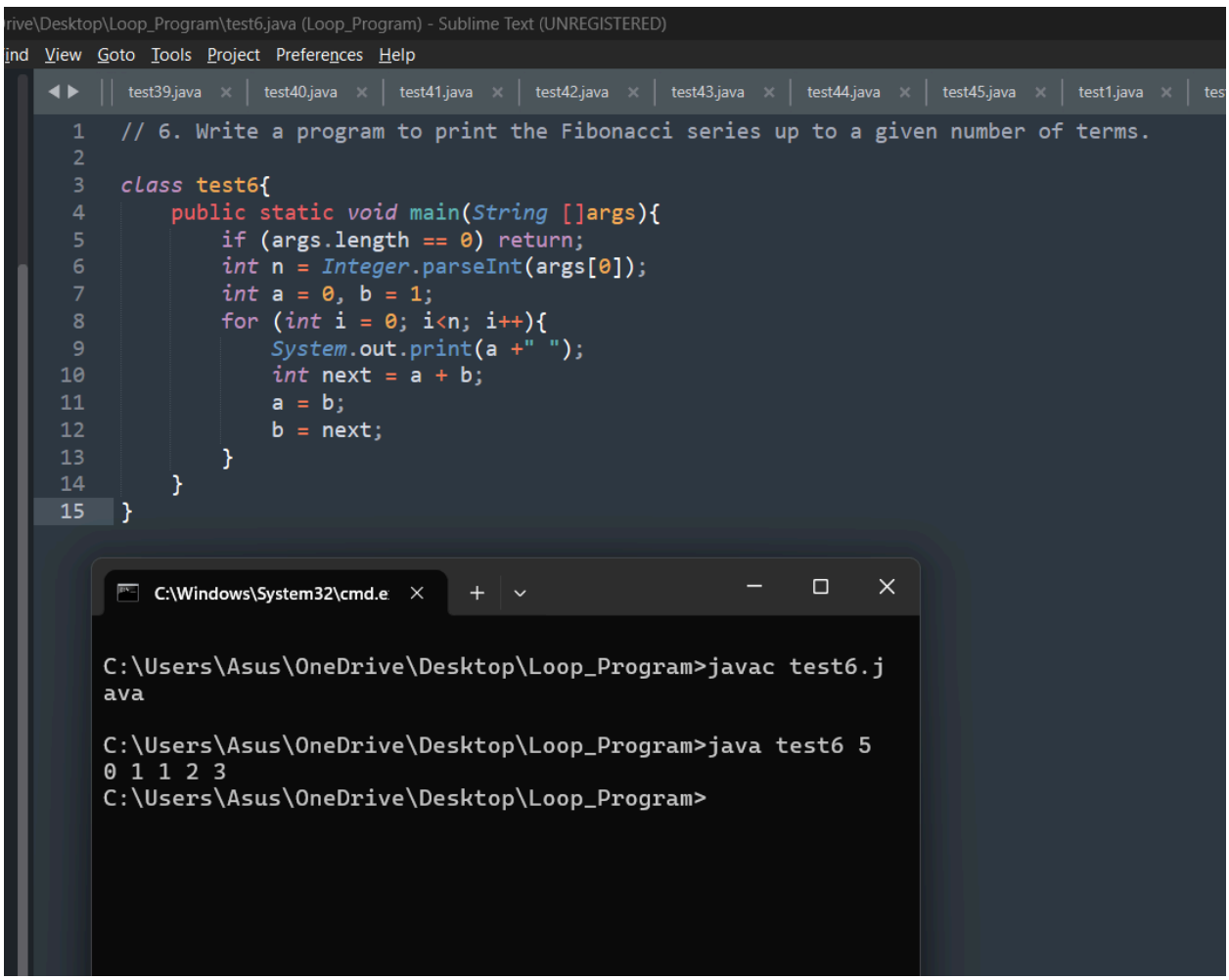
```
class test5{
        public static void main(String []args){
                boolean Isprime = true;
```

```
int n = Integer.parseInt(args[0]);
if (n <=1){
        Isprime = false;
}else{
        for (int i = 2; i * i <= n; i++){
                if(n % 2 == 0){
                        Isprime = false;
                        break;
                }
        }
}
        if (Isprime){
                System.out.println(n +" It's a prime number.");
        }else{
                System.out.println(n + "It's not prime number.");
        }
        }
        }
```



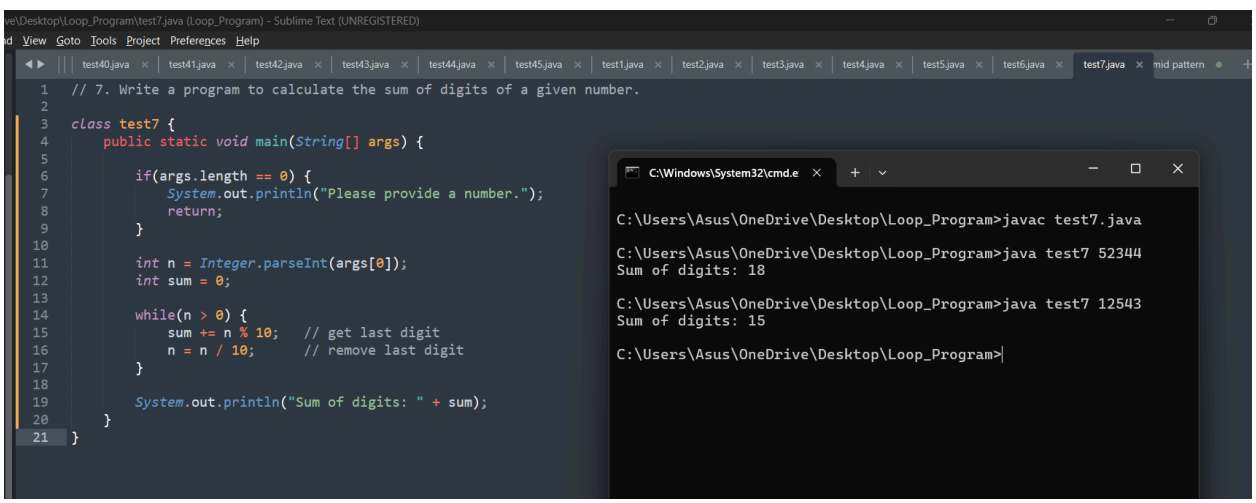
// 6. Write a program to print the Fibonacci series up to a given number of terms.
Solution:

```
class test6{
        public static void main(String []args){
                if (args.length == 0) return;
                int n = Integer.parseInt(args[0]);
                int a = 0, b = 1;
                for (int i = 0; i<n; i++){
                        System.out.print(a +" ");
                        int next = a + b;
                        a = b;
                        b = next;
                }
        }
}
```



// 7. Write a program to calculate the sum of the digits of a given number.
Solution:

```
class test7 {
   public static void main(String[] args) {

      if(args.length == 0) {
         System.out.println("Please provide a number.");
         return;
      }

      int n = Integer.parseInt(args[0]);
      int sum = 0;

      while(n > 0) {
         sum += n % 10;   // get last digit
         n = n / 10;      // remove last digit
      }

      System.out.println("Sum of digits: " + sum);
   }
}
```

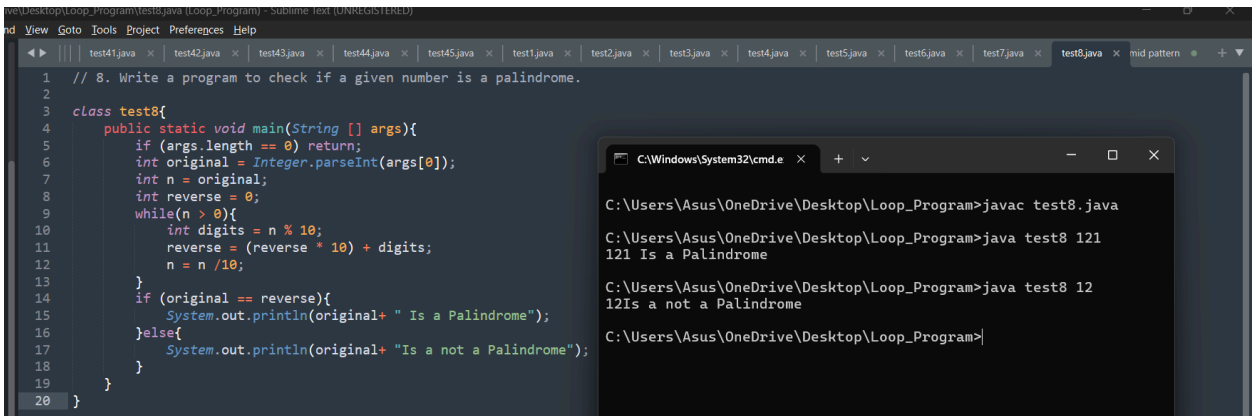


// 8. Write a program to check if a given number is a palindrome.

Solution:

```
class test8{
        public static void main(String [] args){
                if (args.length == 0) return;
                int original = Integer.parseInt(args[0]);
```

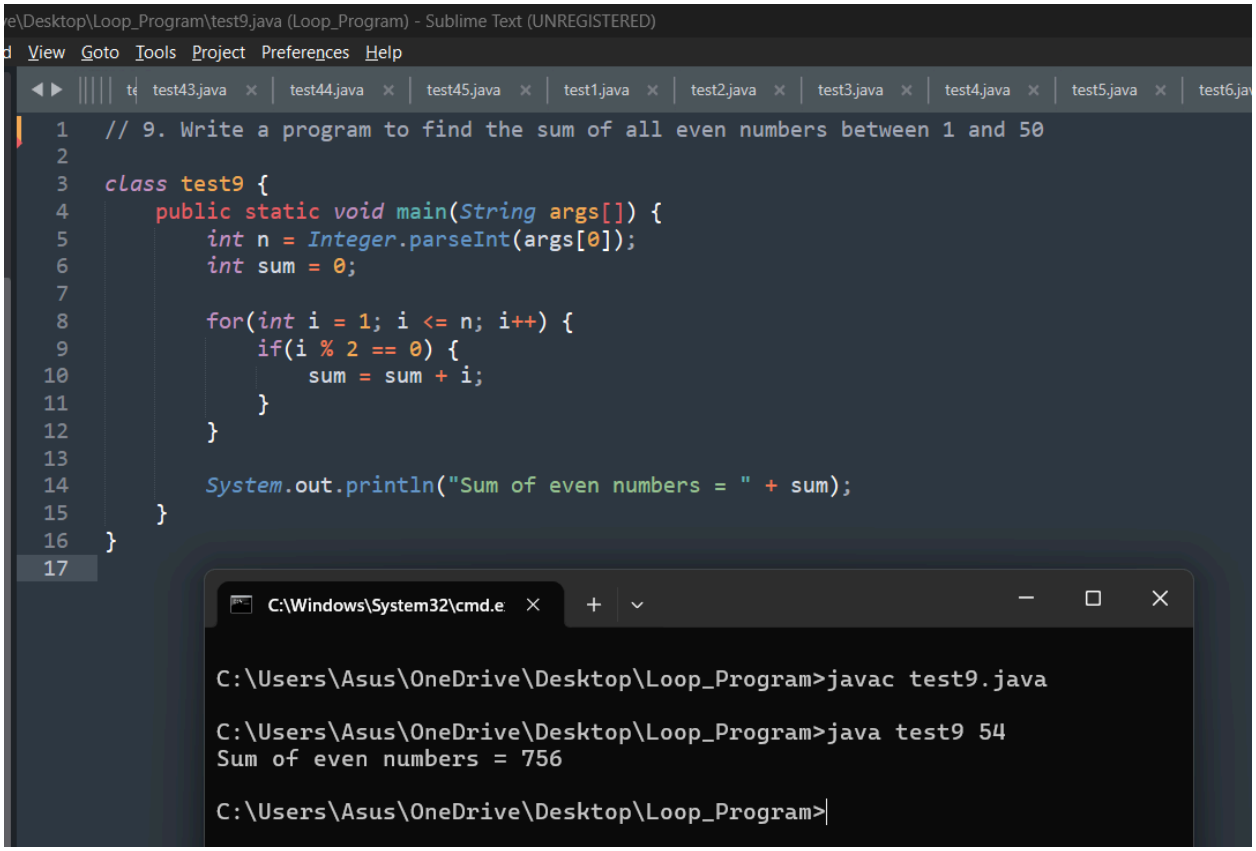
```
            int n = original;
            int reverse = 0;
            while(n > 0){
                int digits = n % 10;
                reverse = (reverse * 10) + digits;
                n = n /10;
            }
            if (original == reverse){
                System.out.println(original+ " Is a Palindrome");
            }else{
                System.out.println(original+ "Is a not a Palindrome");
            }
        }
}
```



// 9. Write a program to find the sum of all even numbers between 1 and 50
Solution:

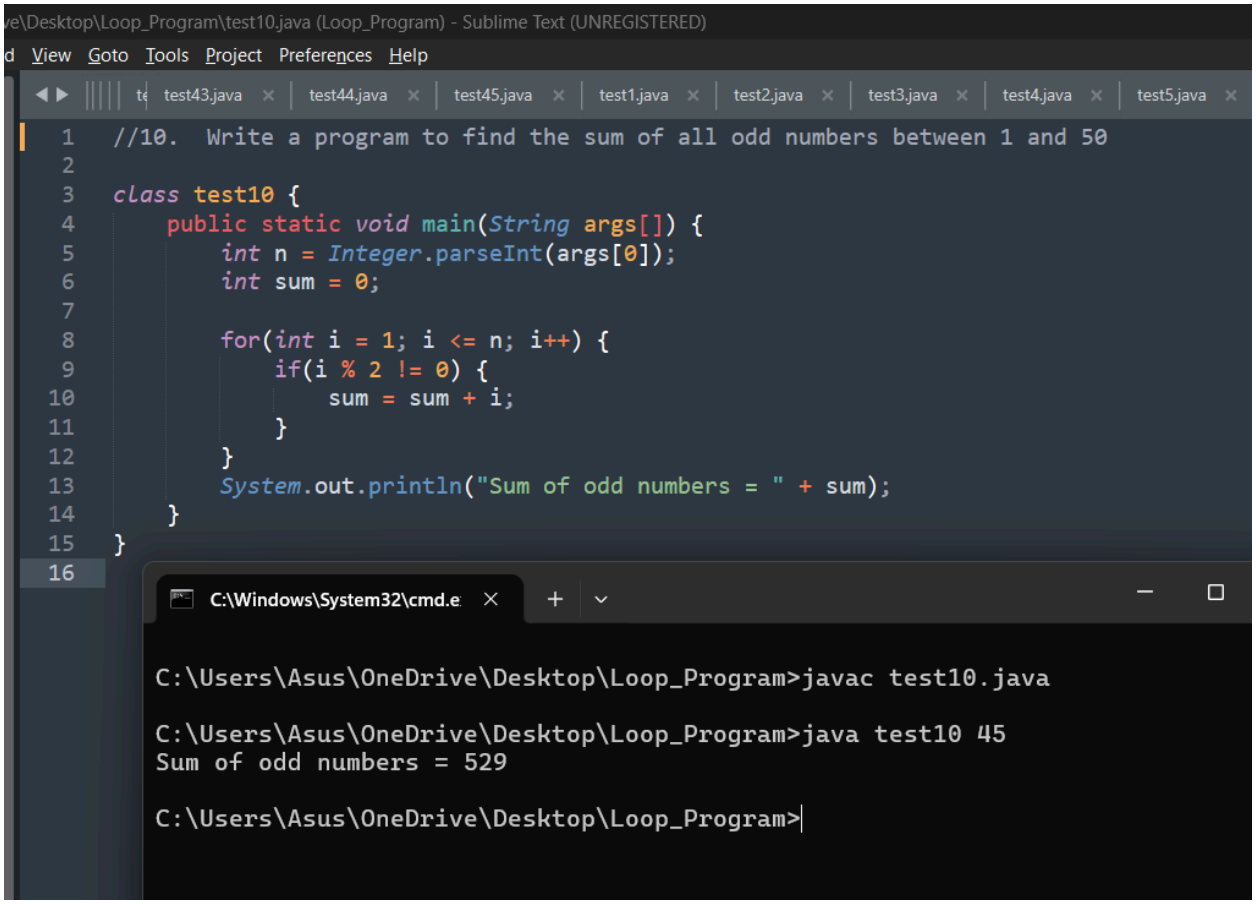
```
class test9 {
    public static void main(String args[]) {
        int n = Integer.parseInt(args[0]);
        int sum = 0;

        for(int i = 1; i <= n; i++) {
            if(i % 2 == 0) {
                sum = sum + i;
            }
        }

        System.out.println("Sum of even numbers = " + sum);
    }
```

}



//10. Write a program to find the sum of all odd numbers between 1 and 50
Solution:

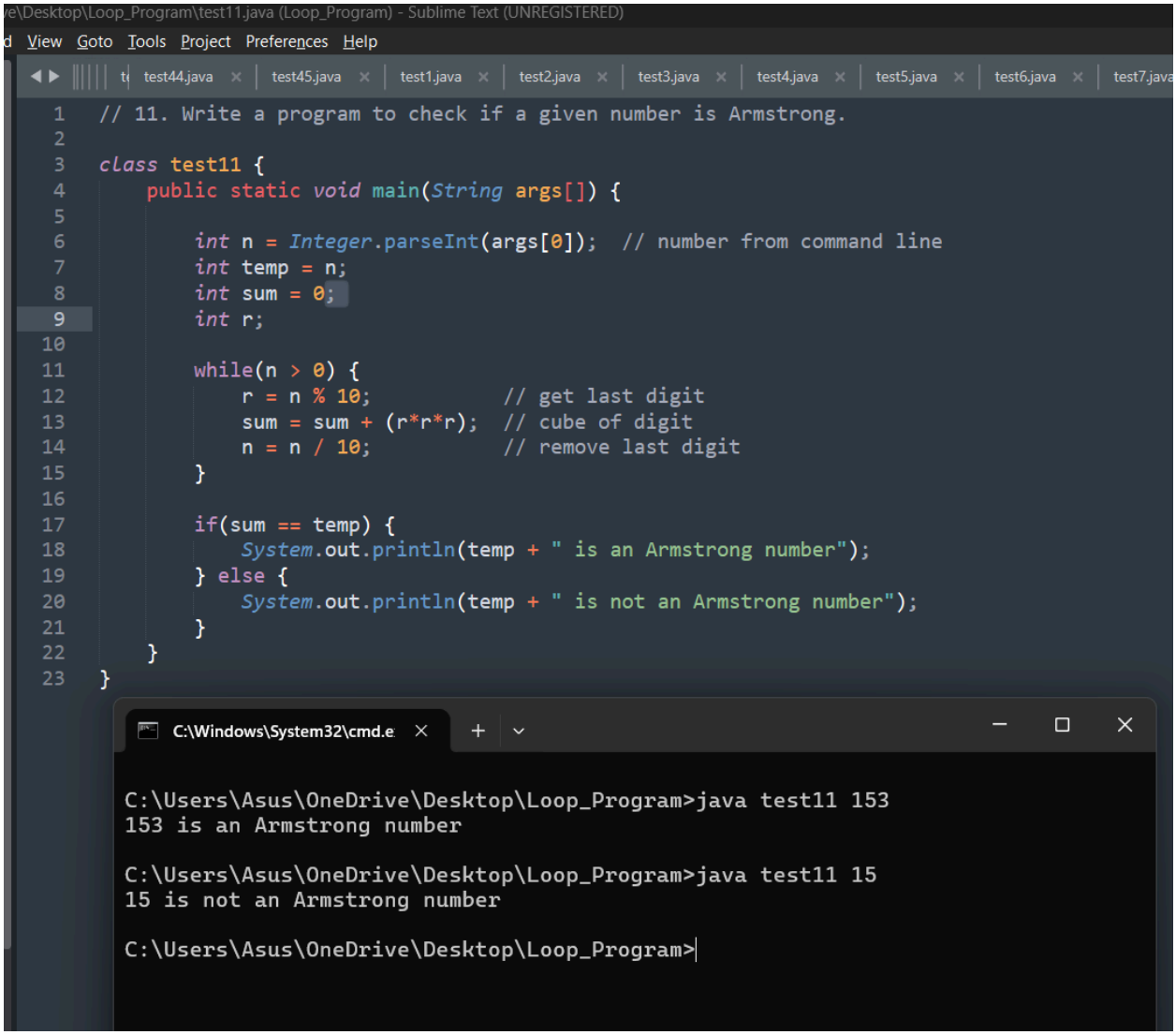
```
class test10 {
   public static void main(String args[]) {
      int n = Integer.parseInt(args[0]);
      int sum = 0;

      for(int i = 1; i <= n; i++) {
         if(i % 2 != 0) {
            sum = sum + i;
         }
      }
      System.out.println("Sum of odd numbers = " + sum);
   }
}
```



// 11. Write a program to check if a given number is Armstrong.
Solution:

```
class test11 {
   public static void main(String args[]) {

      int n = Integer.parseInt(args[0]);  // number from command line
      int temp = n;
      int sum = 0;
      int r;

      while(n > 0) {
         r = n % 10;          // get last digit
         sum = sum + (r*r*r);  // cube of digit
         n = n / 10;          // remove last digit
      }

      if(sum == temp) {
         System.out.println(temp + " is an Armstrong number");
      } else {
```

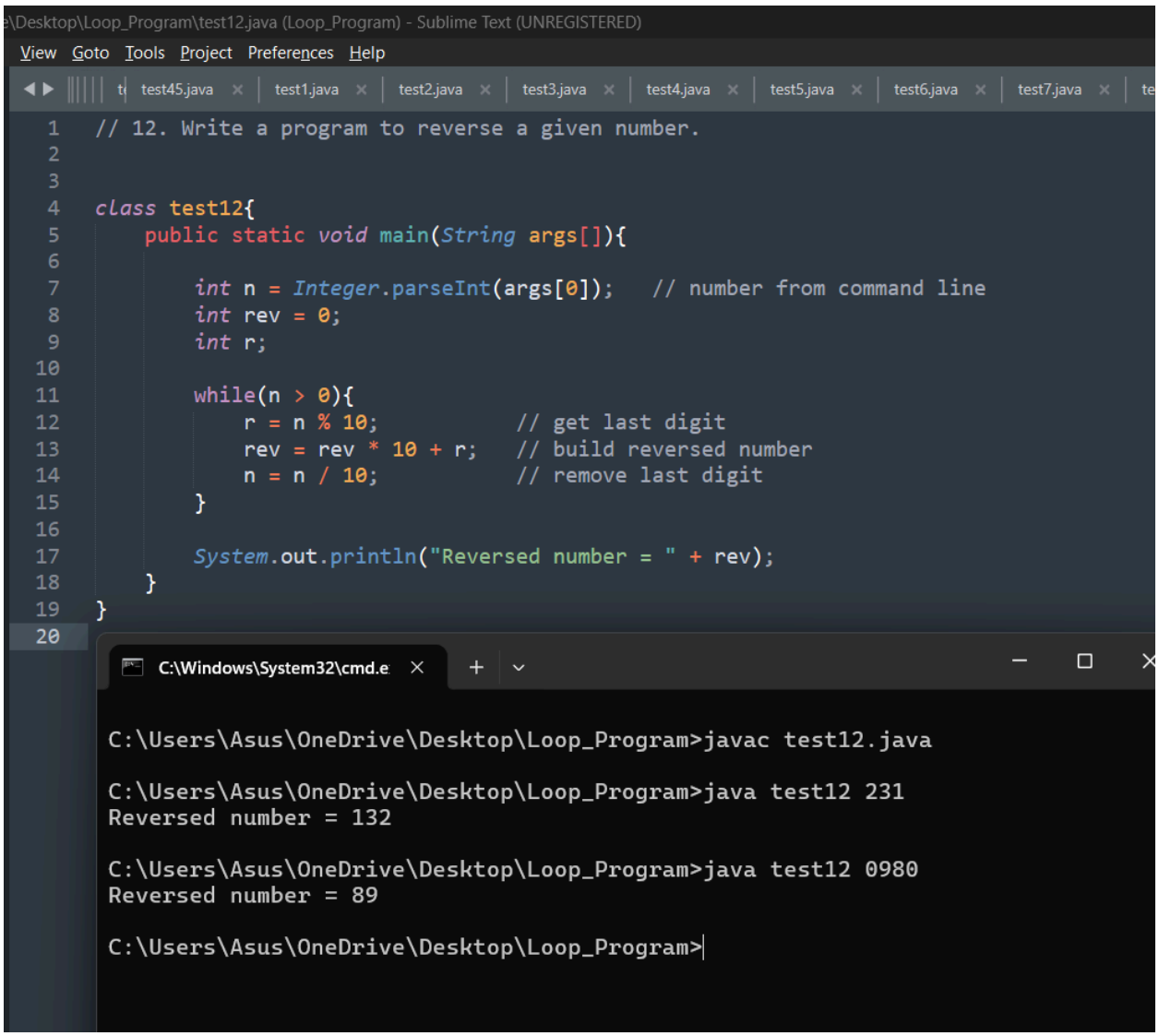
```
            System.out.println(temp + " is not an Armstrong number");
        }
    }
}
```



// 12. Write a program to reverse a given number.
Solution:

```
class test12{
    public static void main(String args[]){

        int n = Integer.parseInt(args[0]);   // number from command line
```

```
        int rev = 0;
        int r;

        while(n > 0){
            r = n % 10;          // get last digit
            rev = rev * 10 + r;   // build reversed number
            n = n / 10;          // remove last digit
        }

        System.out.println("Reversed number = " + rev);
    }
}
```



// 13. Write a program to calculate the power of a number using a loop.

Solution

```
class test13{
   public static void main(String args[]){

      int base = Integer.parseInt(args[0]);   // base number
      int power = Integer.parseInt(args[1]);  // exponent
      int result = 1;

      for(int i = 1; i <= power; i++){
         result = result * base;
      }

      System.out.println("Power = " + result);
   }
}
```

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t test1.java test2.java test3.java test4.java test5.java test6.java test7.java test8.java

```java
// 13. Write a program to calculate the power of a number using a loop.

class test13{
    public static void main(String args[]){

        int base = Integer.parseInt(args[0]);   // base number
        int power = Integer.parseInt(args[1]);  // exponent
        int result = 1;

        for(int i = 1; i <= power; i++){
            result = result * base;
        }

        System.out.println("Power = " + result);
    }
}
```

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```
C:\Users\Asus\OneDrive\Desktop\Loop_Program>javac test13.java

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test13 21 1
Power = 21

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test13 21 2
Power = 441

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test13 4 2
Power = 16

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test13 4 3
Power = 64

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```

// 14. Write a program to find the greatest common divisor (GCD) of two numbers.
Solution:

```
class test14{
   public static void main(String args[]){

      int a = Integer.parseInt(args[0]);
      int b = Integer.parseInt(args[1]);
```

```
        int gcd = 1;

        for(int i = 1; i <= a && i <= b; i++){
            if(a % i == 0 && b % i == 0){
                gcd = i;
            }
        }

        System.out.println("GCD = " + gcd);
    }
}
```

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test2.java | test3.java | test4.java | test5.java | test6.java | test7.java | test8.java | test9.java

```
// 14. Write a program to find the greatest common divisor (GCD) of two numbers.

class test14{
    public static void main(String args[]){

        int a = Integer.parseInt(args[0]);
        int b = Integer.parseInt(args[1]);

        int gcd = 1;

        for(int i = 1; i <= a && i <= b; i++){
            if(a % i == 0 && b % i == 0){
                gcd = i;
            }
        }

        System.out.println("GCD = " + gcd);
    }
}
```

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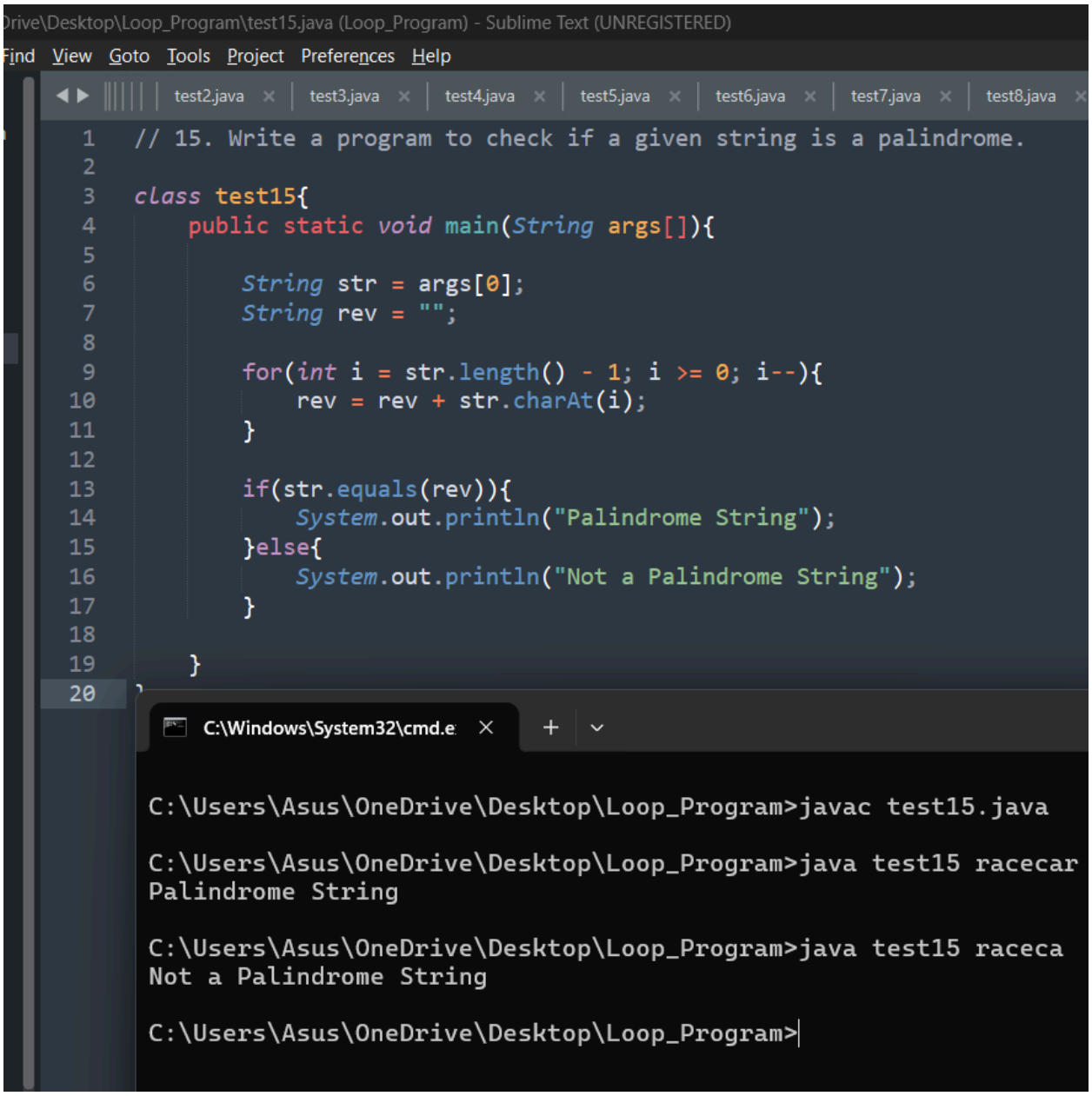
```
C:\Users\Asus\OneDrive\Desktop\Loop_Program>javac test14.java

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test14 44 42
GCD = 2

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test14 44 41
GCD = 1

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test14 44 80
GCD = 4

C:\Users\Asus\OneDrive\Desktop\Loop_Program>
```

// 15. Write a program to check if a given string is a palindrome.
Solution:

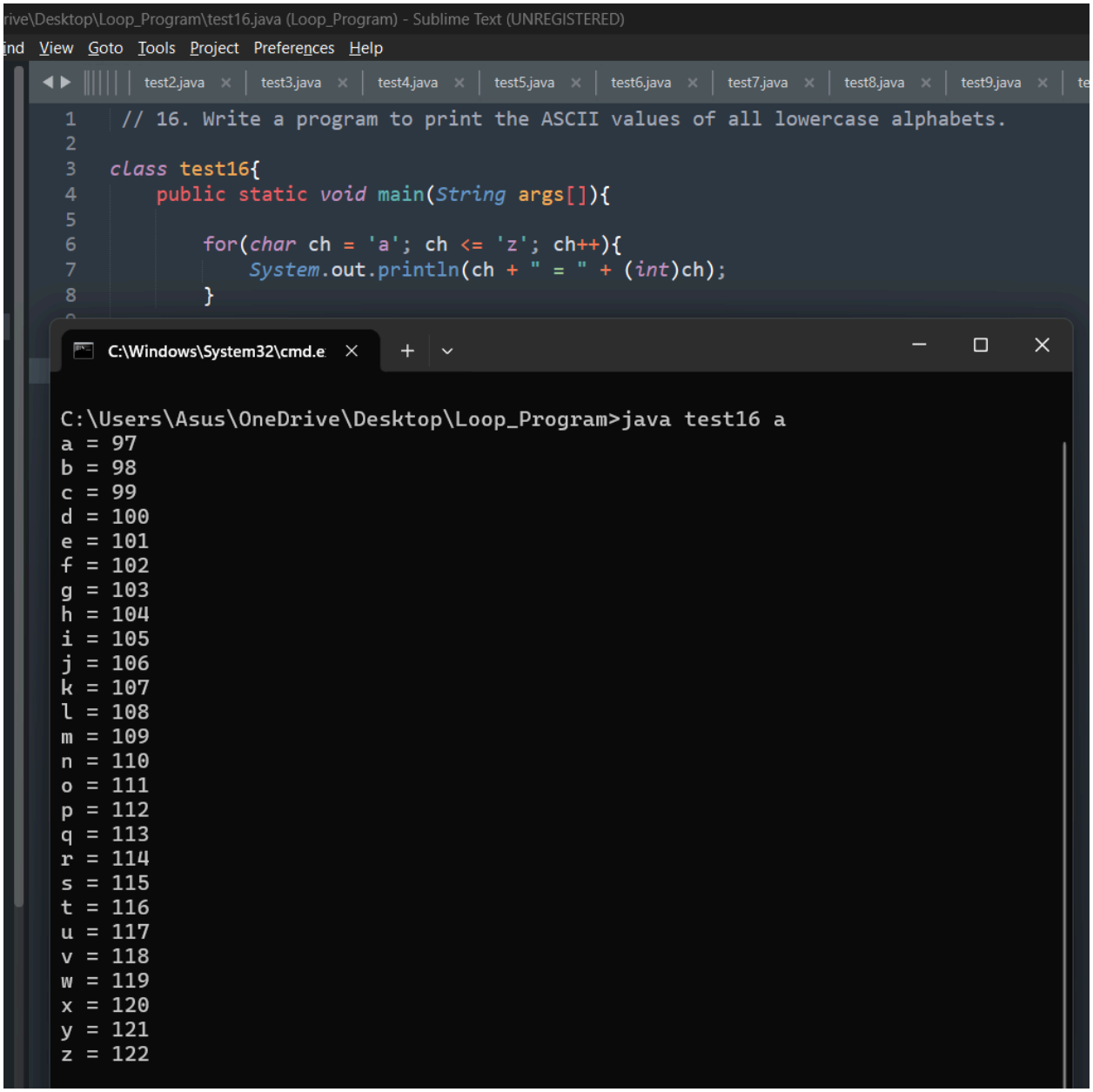
```
class test15{
    public static void main(String args[]){

        String str = args[0];
        String rev = "";

        for(int i = str.length() - 1; i >= 0; i--){
            rev = rev + str.charAt(i);
        }

        if(str.equals(rev)){
            System.out.println("Palindrome String");
        }else{
            System.out.println("Not a Palindrome String");
        }

    }
}
```



// 16. Write a program to print the ASCII values of all lowercase alphabets.
Solution:

```
class test16{
   public static void main(String args[]){

      for(char ch = 'a'; ch <= 'z'; ch++){
         System.out.println(ch + " = " + (int)ch);
      }
```

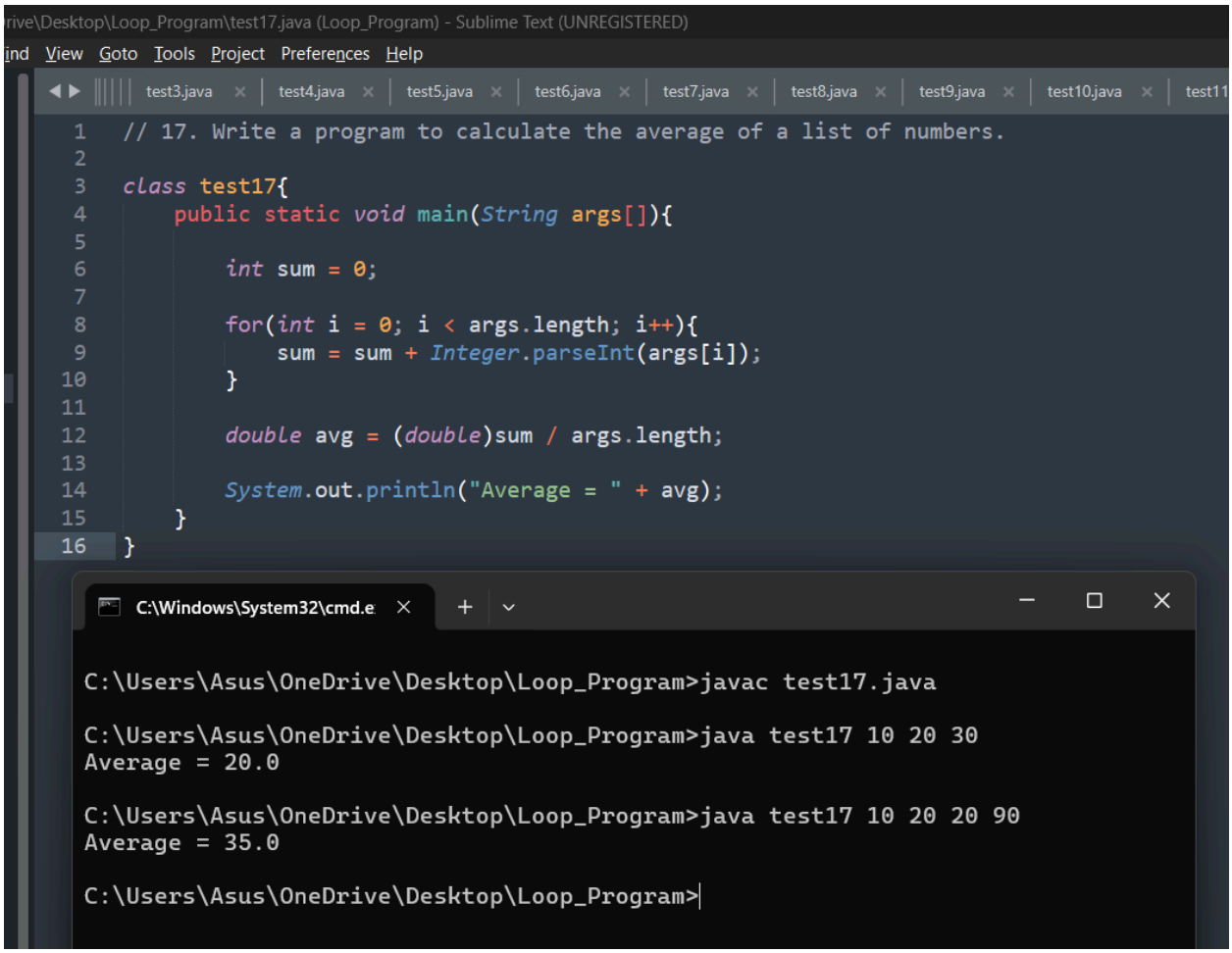
```
    }
}
```



// 17. Write a program to calculate the average of a list of numbers.
Solution:

```
class test17{
    public static void main(String args[]){

        int sum = 0;
```

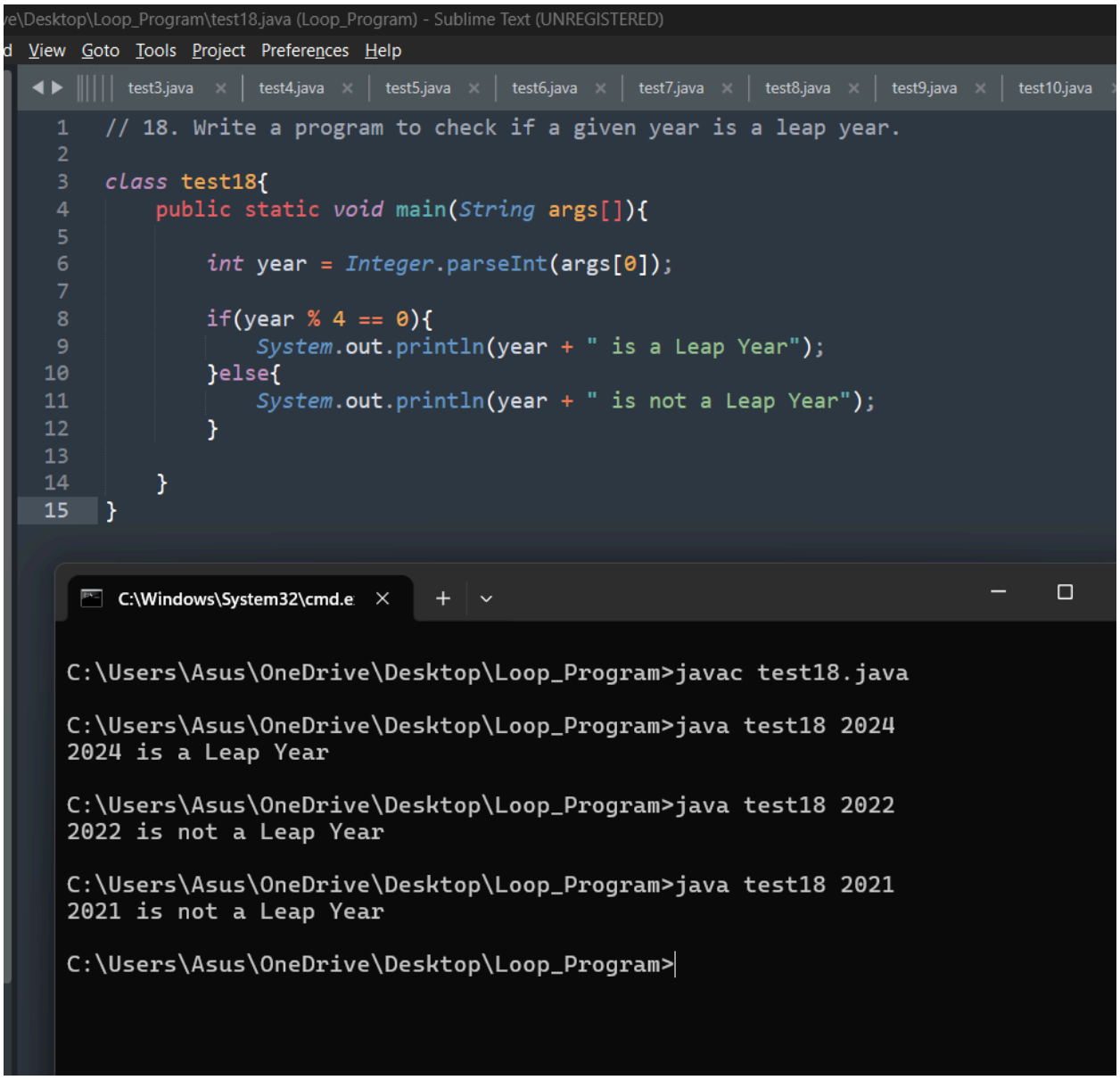
```
        for(int i = 0; i < args.length; i++){
            sum = sum + Integer.parseInt(args[i]);
        }

        double avg = (double)sum / args.length;

        System.out.println("Average = " + avg);
    }
}
```



// 18. Write a program to check if a given year is a leap year.
Solution:

```
class test18{
    public static void main(String args[]){

        int year = Integer.parseInt(args[0]);
```

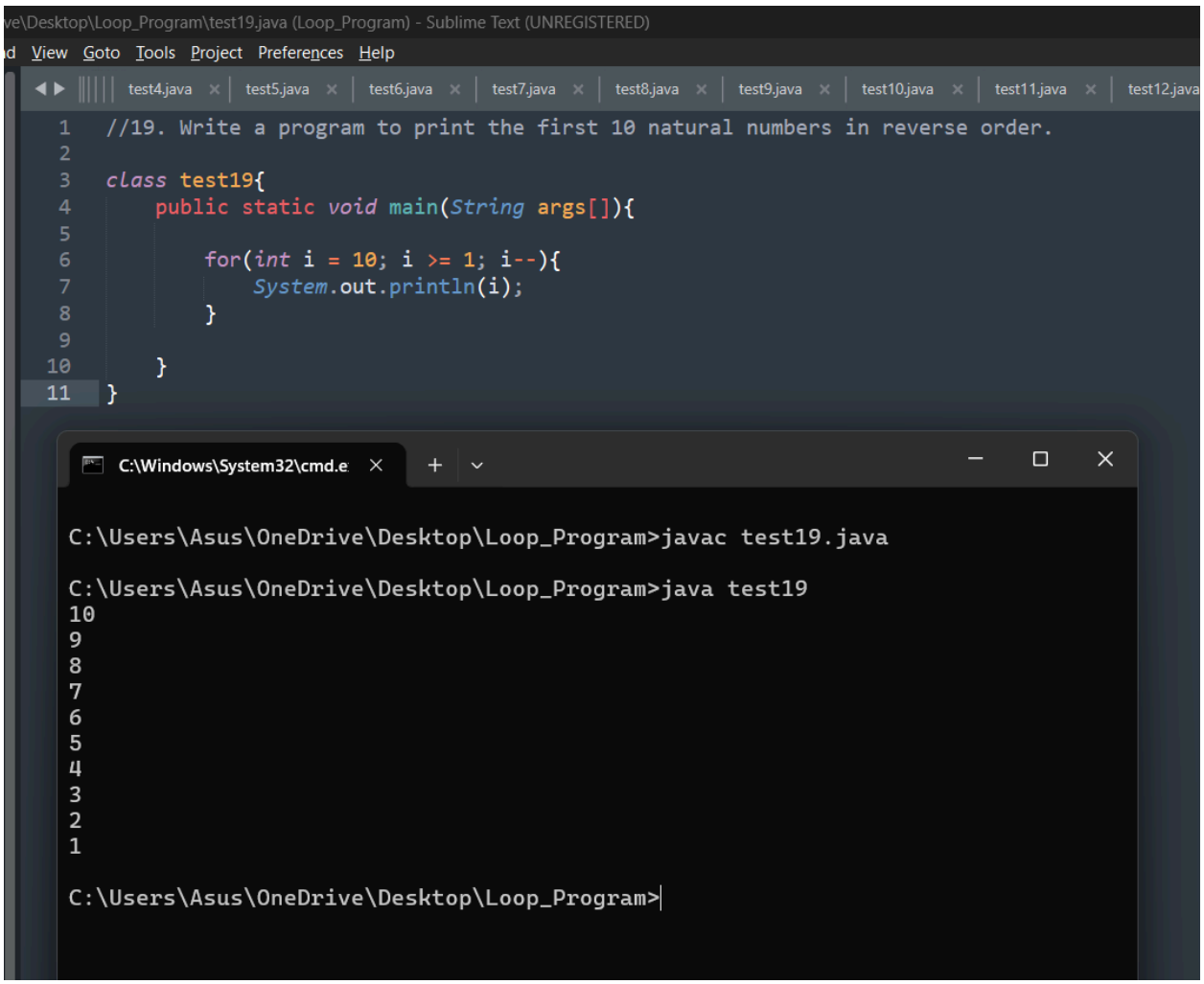
```
        if(year % 4 == 0){
            System.out.println(year + " is a Leap Year");
        }else{
            System.out.println(year + " is not a Leap Year");
        }

    }
}
```



//19. Write a program to print the first 10 natural numbers in reverse order.

Solution:

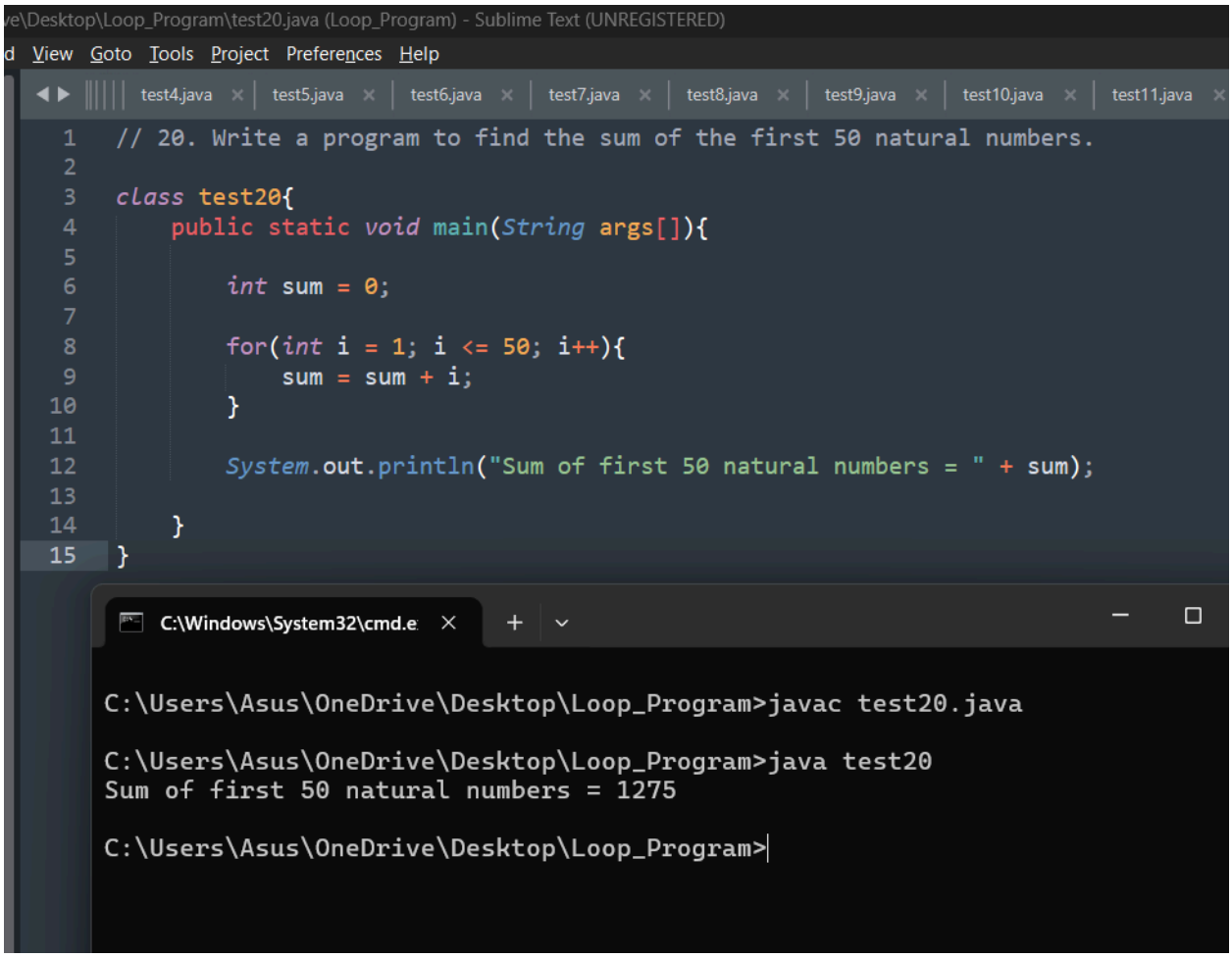
```
class test19{
   public static void main(String args[]){

      for(int i = 10; i >= 1; i--){
         System.out.println(i);
      }

   }
}
```



// 20. Write a program to find the sum of the first 50 natural numbers.
Solution:

```
class test20{
```

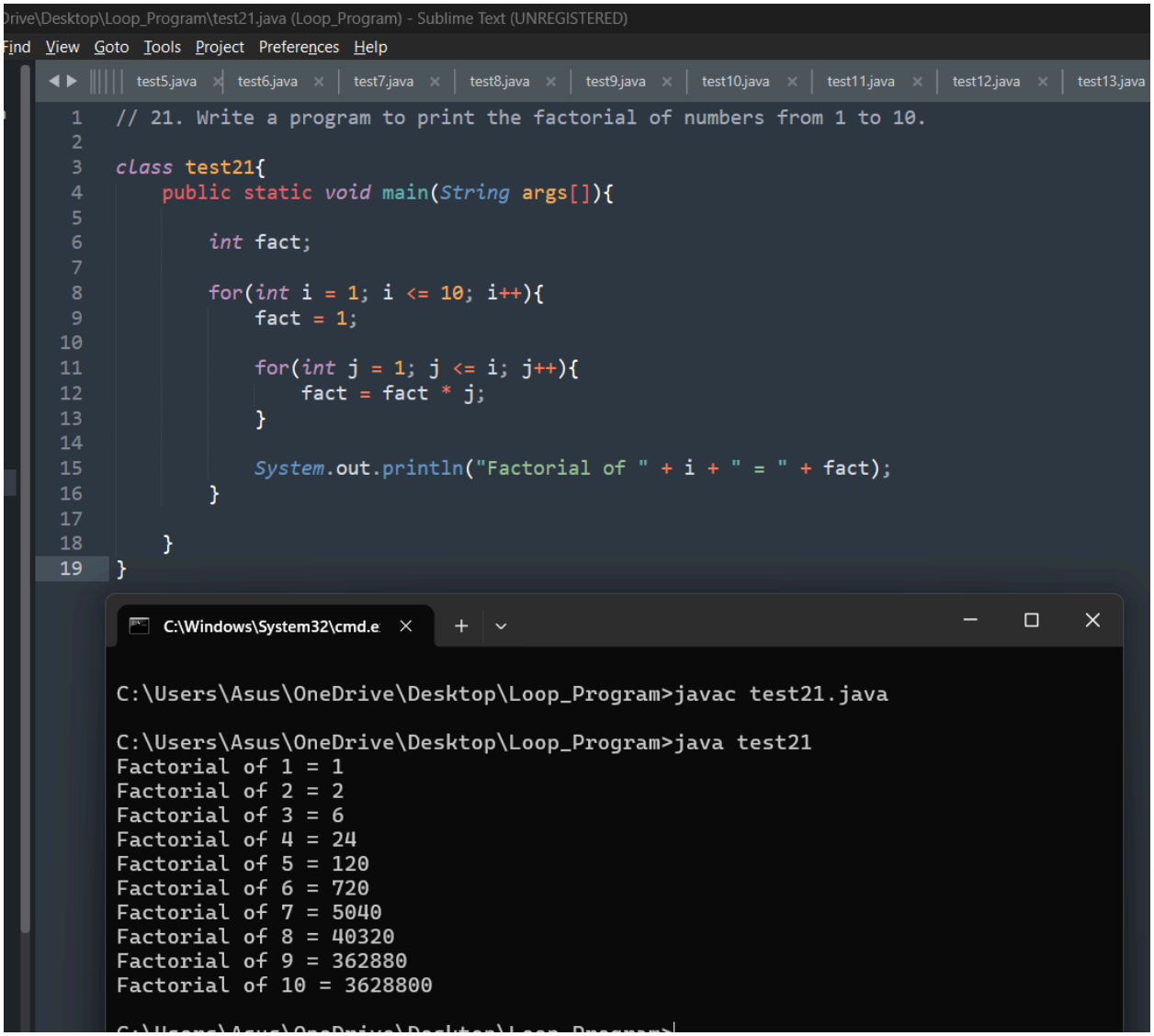
```
    public static void main(String args[]){

        int sum = 0;

        for(int i = 1; i <= 50; i++){
            sum = sum + i;
        }

        System.out.println("Sum of first 50 natural numbers = " + sum);

    }
}
```



// 21. Write a program to print the factorial of numbers from 1 to 10.
Solution:

```
class test21{
    public static void main(String args[]){
```

```
        int fact;

        for(int i = 1; i <= 10; i++){
            fact = 1;

            for(int j = 1; j <= i; j++){
                fact = fact * j;
            }

            System.out.println("Factorial of " + i + " = " + fact);
        }

    }
}
```

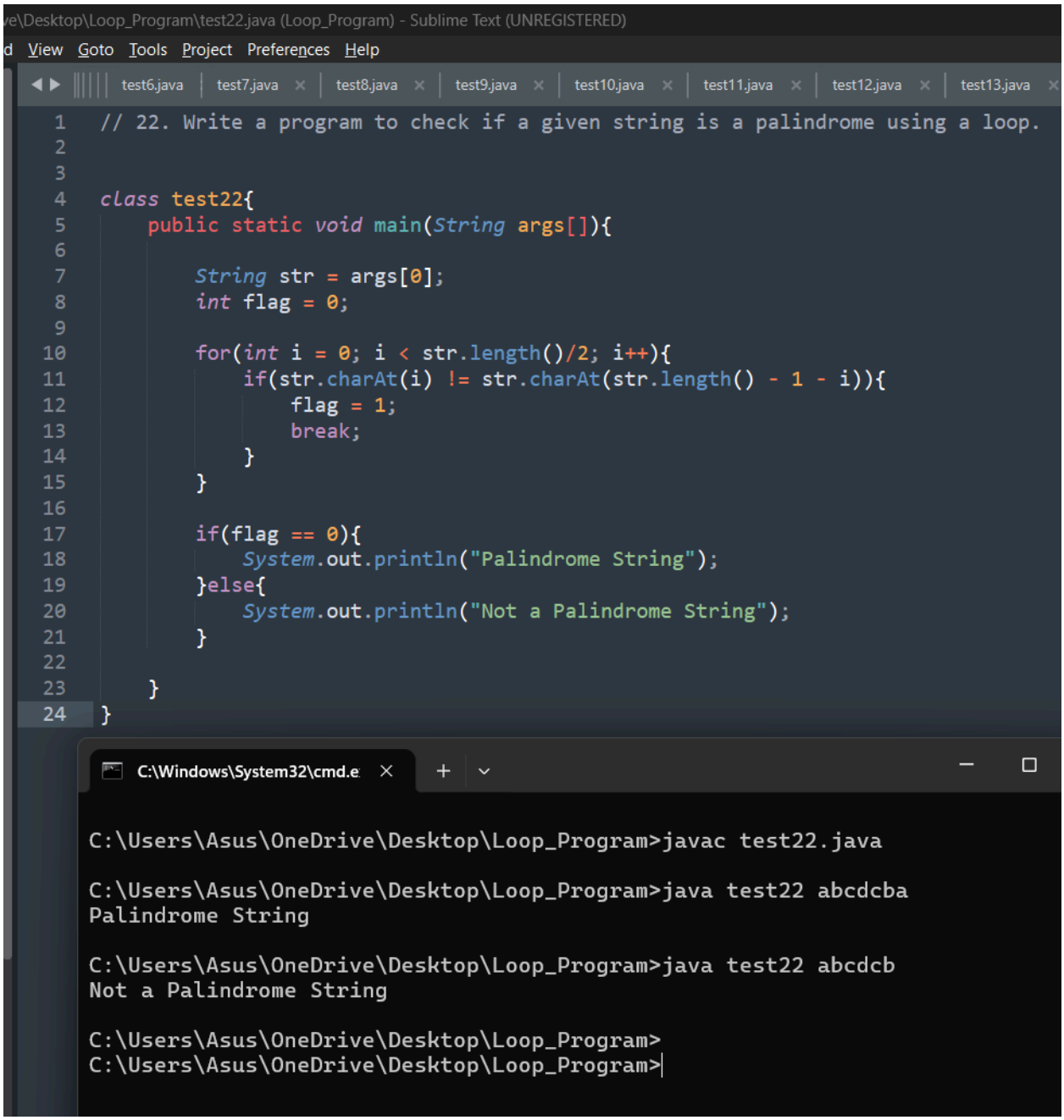


// 22. Write a program to check if a given string is a palindrome using a loop.

Solution:

```
class test22{
   public static void main(String args[]){

      String str = args[0];
      int flag = 0;

      for(int i = 0; i < str.length()/2; i++){
         if(str.charAt(i) != str.charAt(str.length() - 1 - i)){
            flag = 1;
```

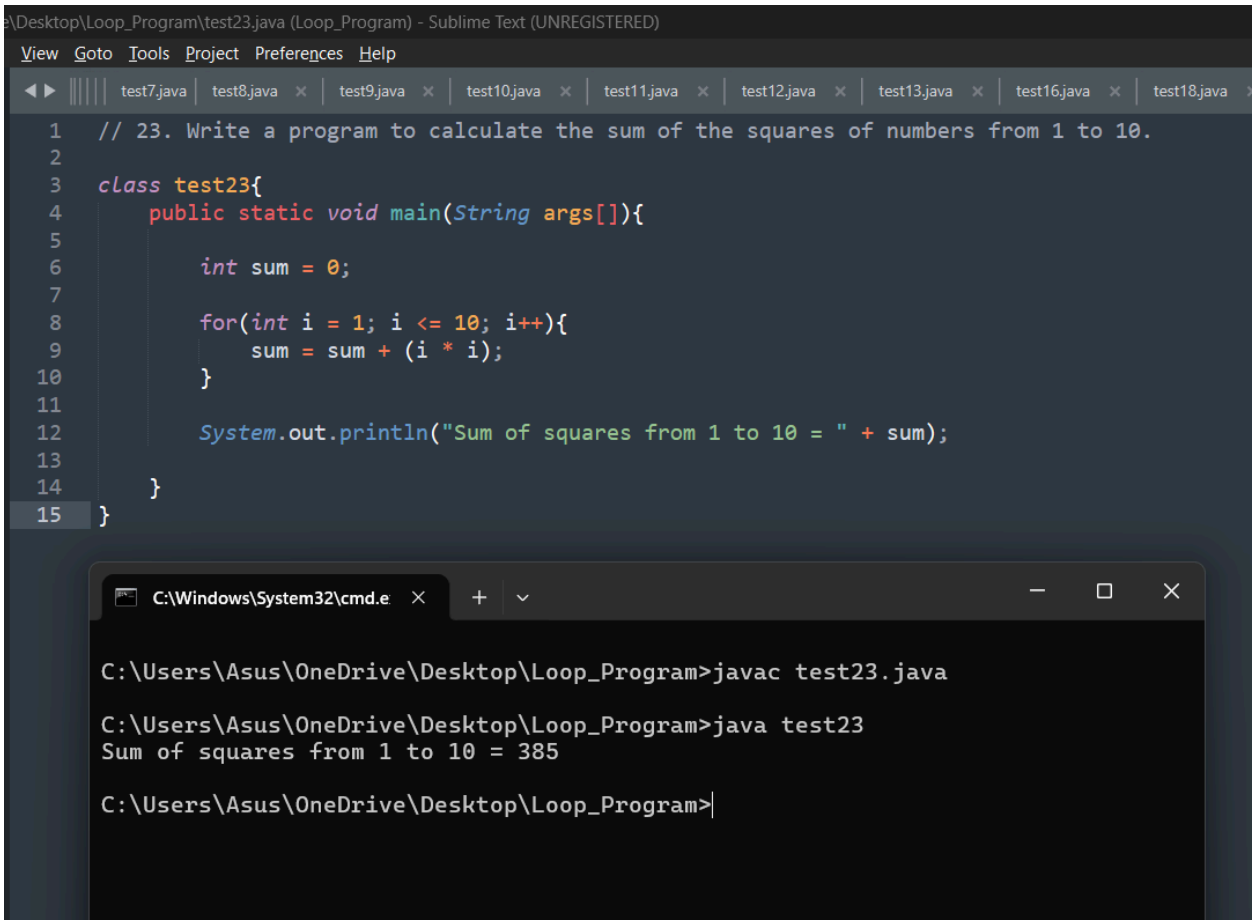
```
            break;
        }
    }

    if(flag == 0){
        System.out.println("Palindrome String");
    }else{
        System.out.println("Not a Palindrome String");
    }

  }
}
```



// 23. Write a program to calculate the sum of the squares of numbers from 1 to 10.
Solution:

```
class test23{
   public static void main(String args[]){

      int sum = 0;
```

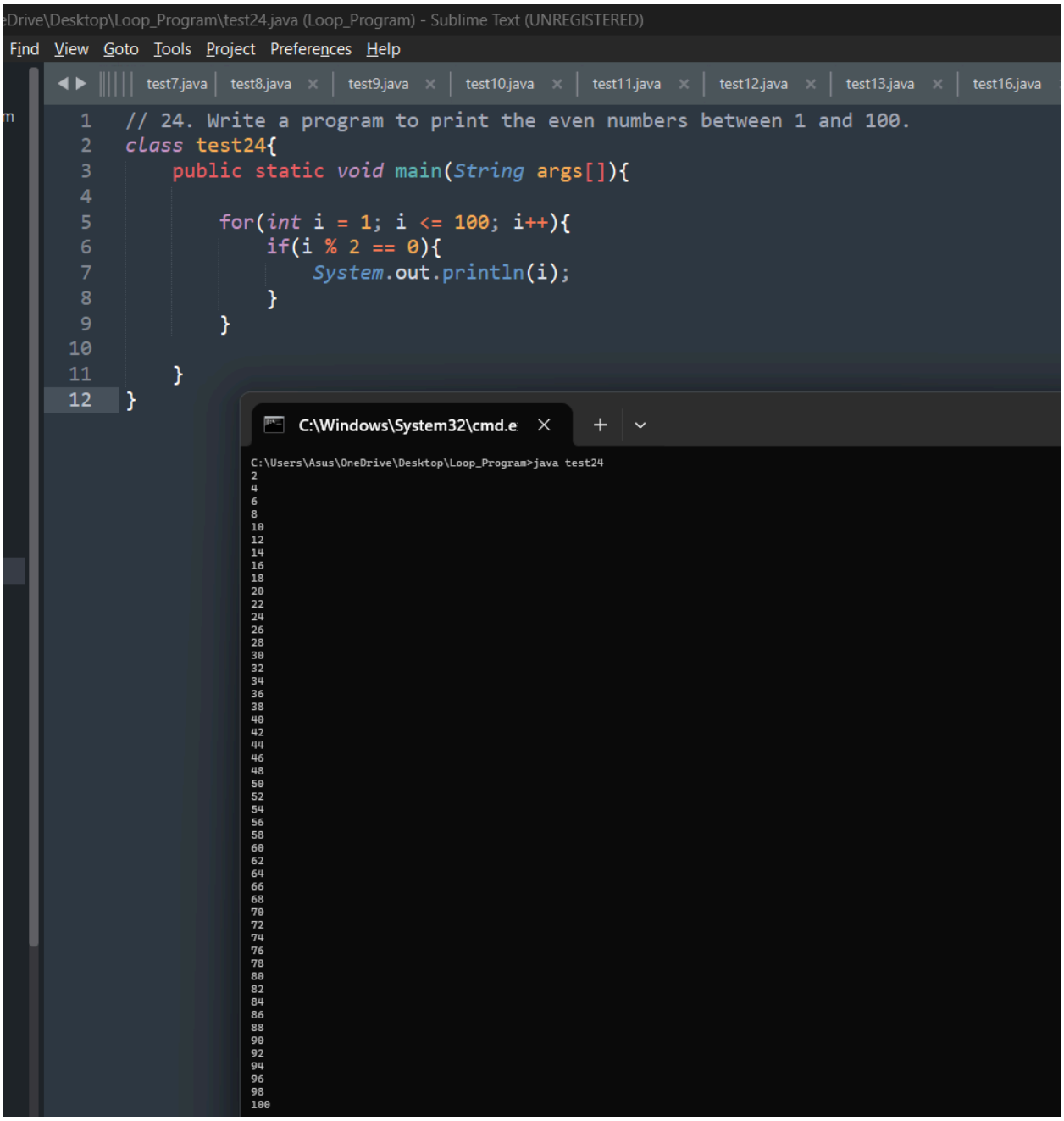
```
        for(int i = 1; i <= 10; i++){
            sum = sum + (i * i);
        }

        System.out.println("Sum of squares from 1 to 10 = " + sum);

    }
}
```



// 24. Write a program to print the even numbers between 1 and 100.
Solution:

```
class test24{
    public static void main(String args[]){

        for(int i = 1; i <= 100; i++){
            if(i % 2 == 0){
                System.out.println(i);
```

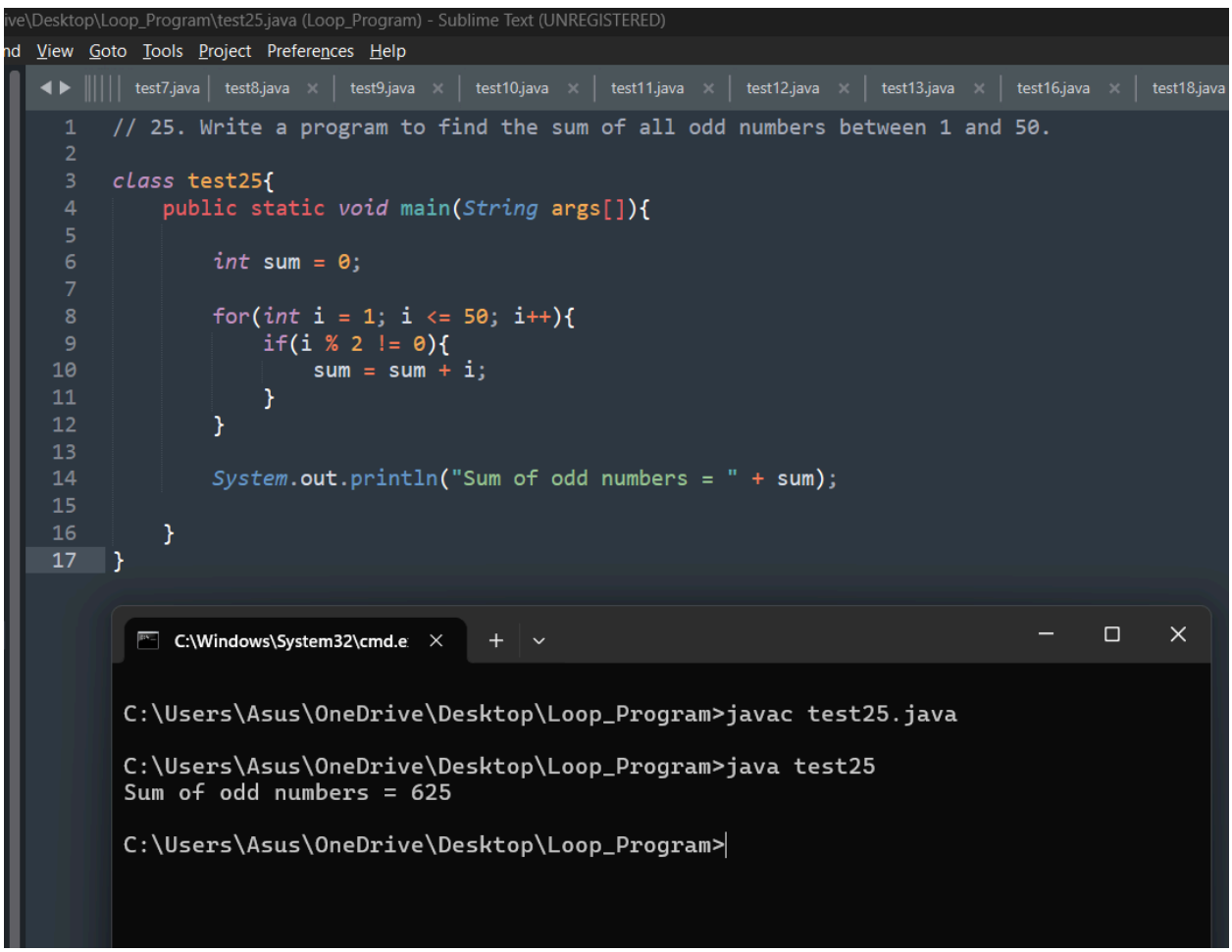
```
        }
    }

  }
}
```



// 25. Write a program to find the sum of all odd numbers between 1 and 50.

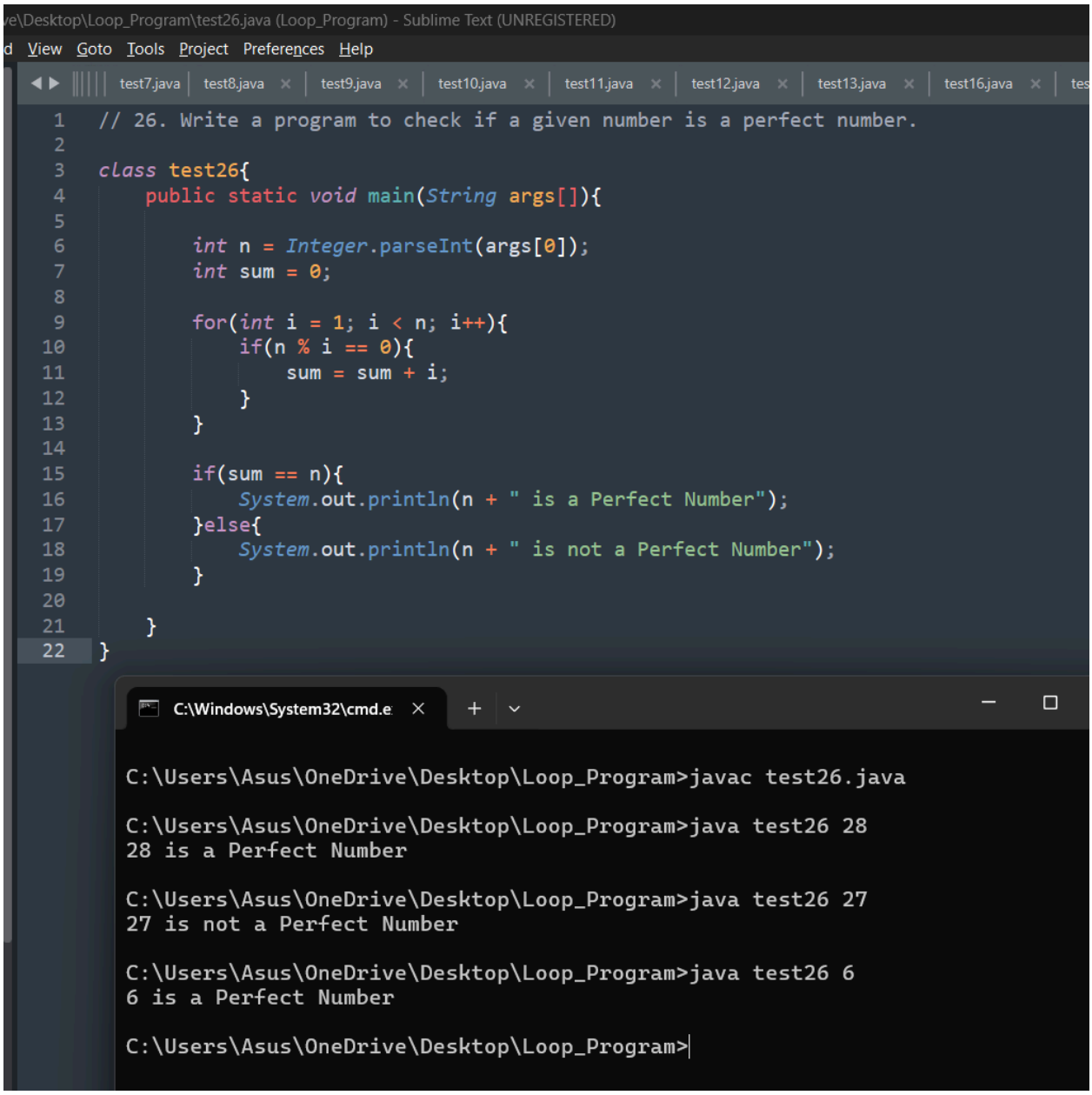
Solution:

```
class test25{
   public static void main(String args[]){

      int sum = 0;

      for(int i = 1; i <= 50; i++){
         if(i % 2 != 0){
            sum = sum + i;
         }
      }

      System.out.println("Sum of odd numbers = " + sum);

   }
}
```



// 26. Write a program to check if a given number is a perfect number.
Solution:

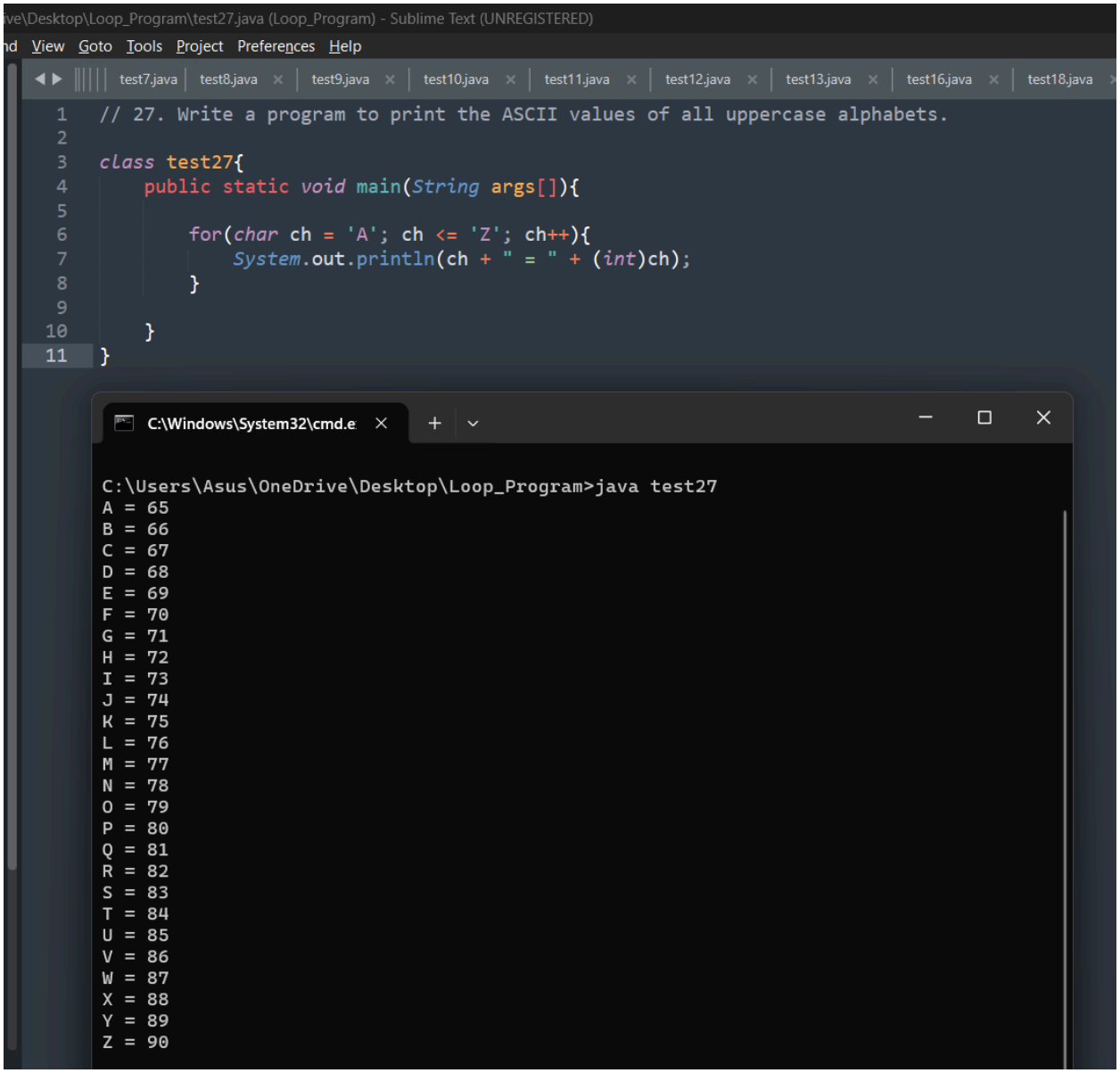
```
class test26{
    public static void main(String args[]){

        int n = Integer.parseInt(args[0]);
        int sum = 0;

        for(int i = 1; i < n; i++){
            if(n % i == 0){
                sum = sum + i;
            }
        }

        if(sum == n){
            System.out.println(n + " is a Perfect Number");
        }else{
            System.out.println(n + " is not a Perfect Number");
        }

    }
}
```



// 27. Write a program to print the ASCII values of all uppercase alphabets.
Solution:

```
class test27{
   public static void main(String args[]){

      for(char ch = 'A'; ch <= 'Z'; ch++){
         System.out.println(ch + " = " + (int)ch);
      }

   }
```

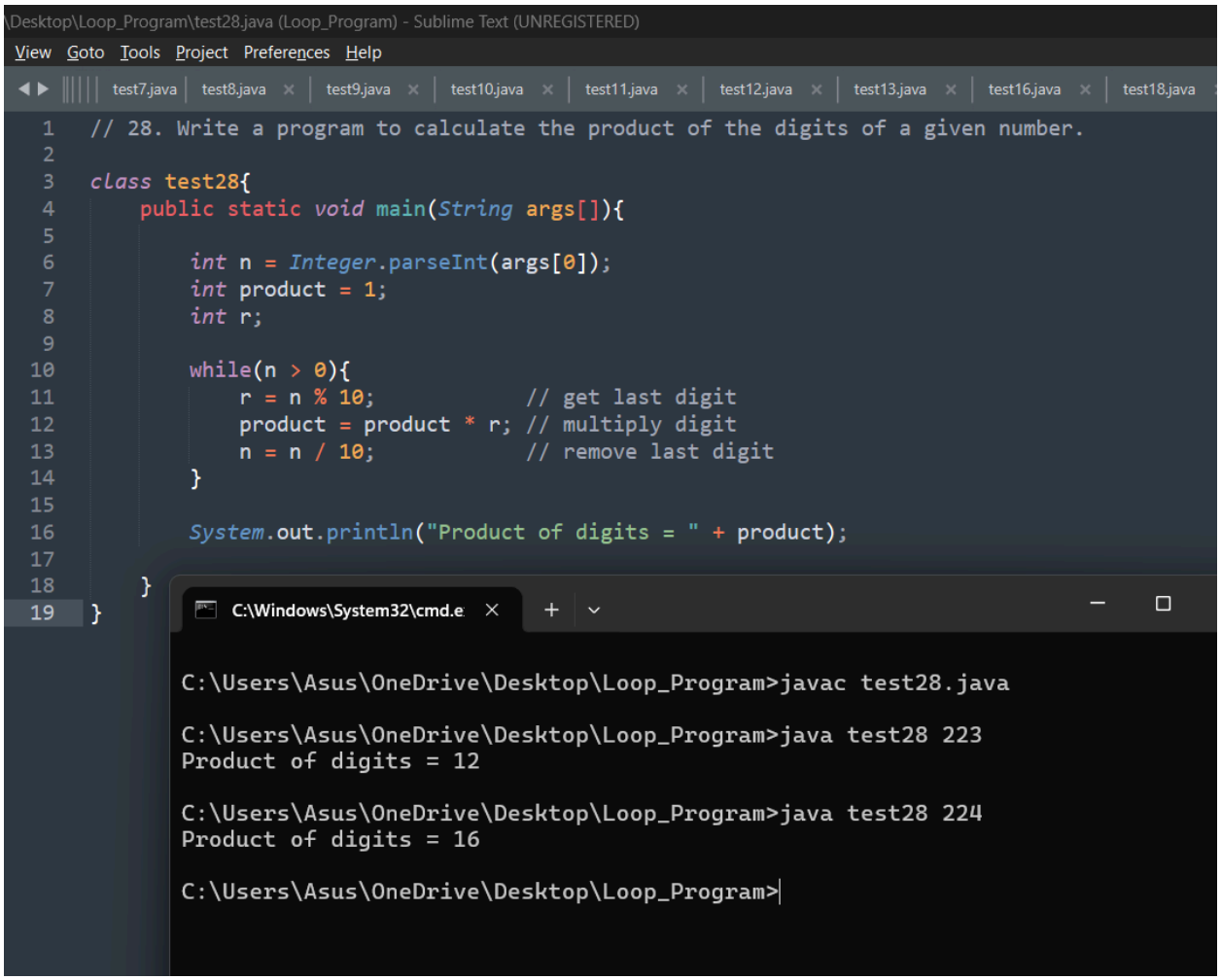
}



// 28. Write a program to calculate the product of the digits of a given number.
Solution:

```
class test28{
   public static void main(String args[]){

      int n = Integer.parseInt(args[0]);
      int product = 1;
      int r;
```

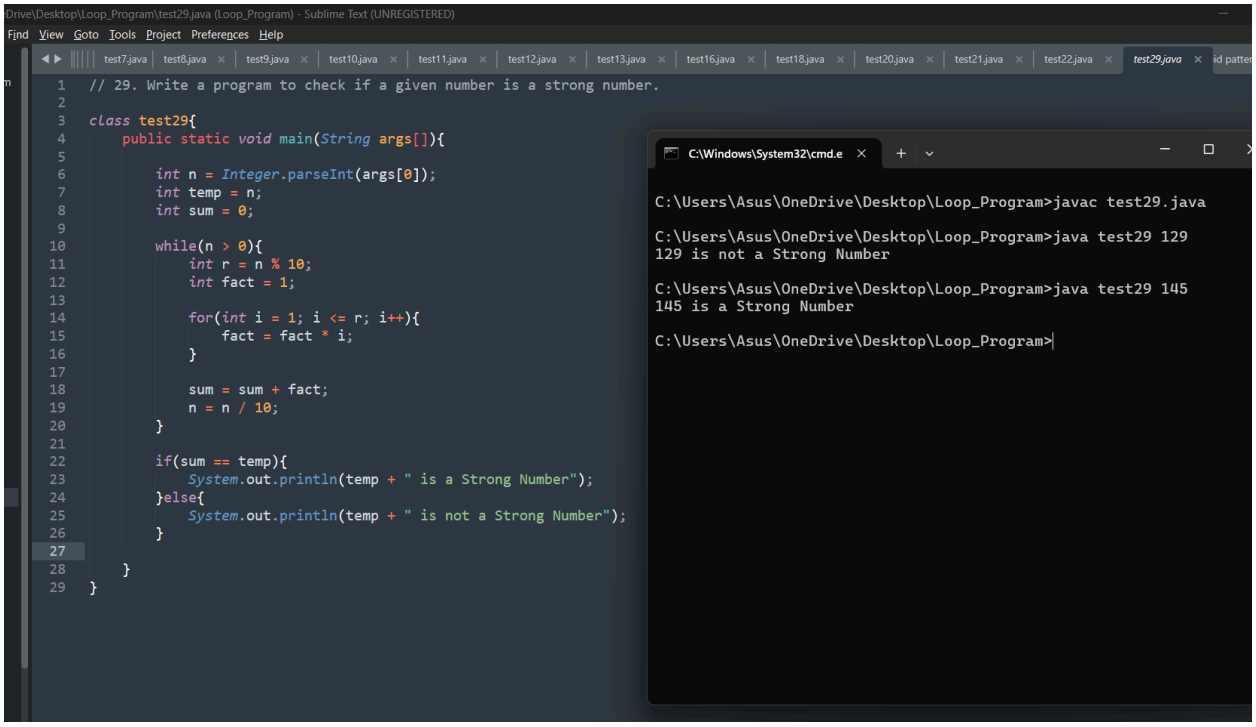
```
        while(n > 0){
            r = n % 10;            // get last digit
            product = product * r; // multiply digit
            n = n / 10;            // remove last digit
        }

        System.out.println("Product of digits = " + product);

    }
}
```



// 29. Write a program to check if a given number is a strong number.
Solution:

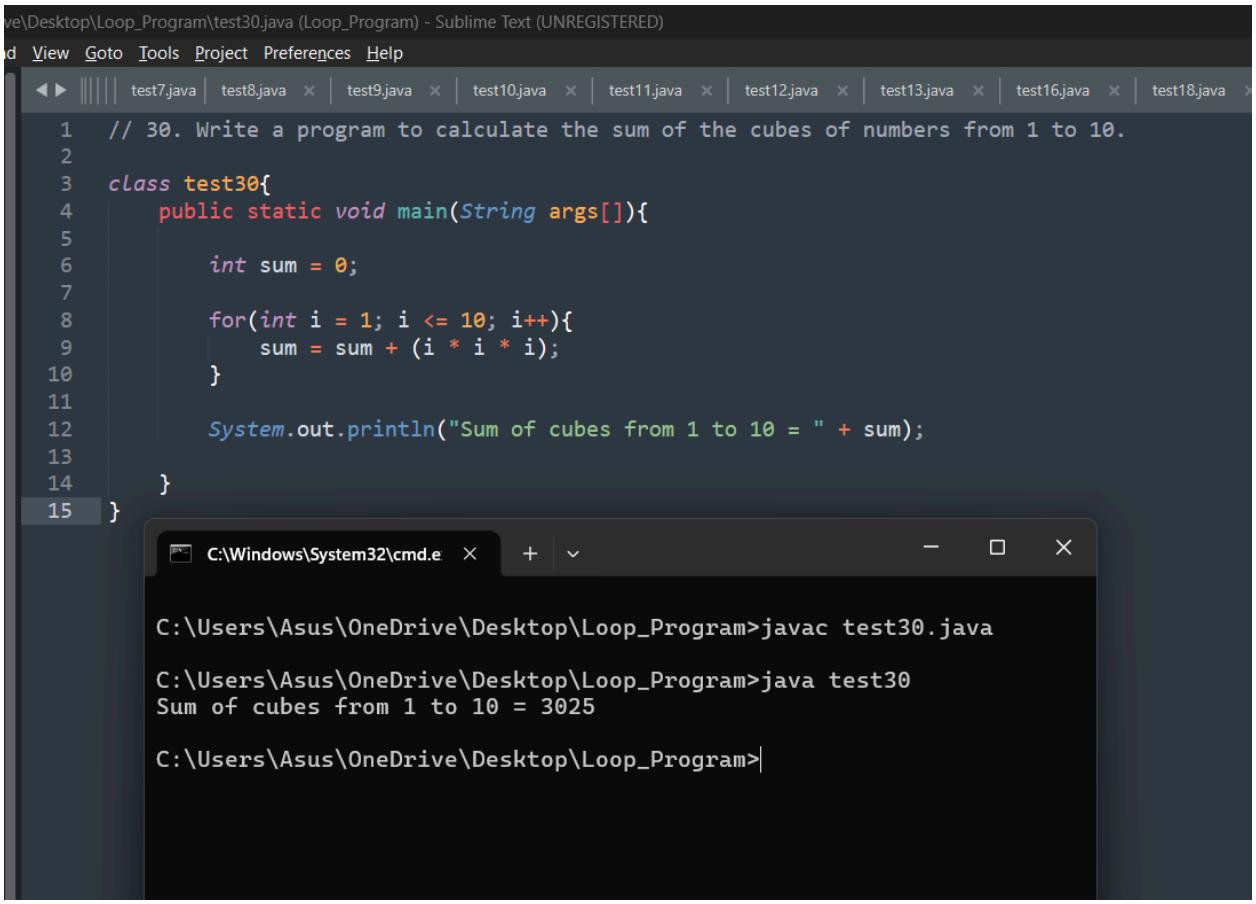
```
class test29{
    public static void main(String args[]){
```

```
        int n = Integer.parseInt(args[0]);
        int temp = n;
        int sum = 0;

        while(n > 0){
            int r = n % 10;
            int fact = 1;

            for(int i = 1; i <= r; i++){
                fact = fact * i;
            }

            sum = sum + fact;
            n = n / 10;
        }

        if(sum == temp){
            System.out.println(temp + " is a Strong Number");
        }else{
            System.out.println(temp + " is not a Strong Number");
        }

    }
}
```



// 30. Write a program to calculate the sum of the cubes of numbers from 1 to 10.
Solution:

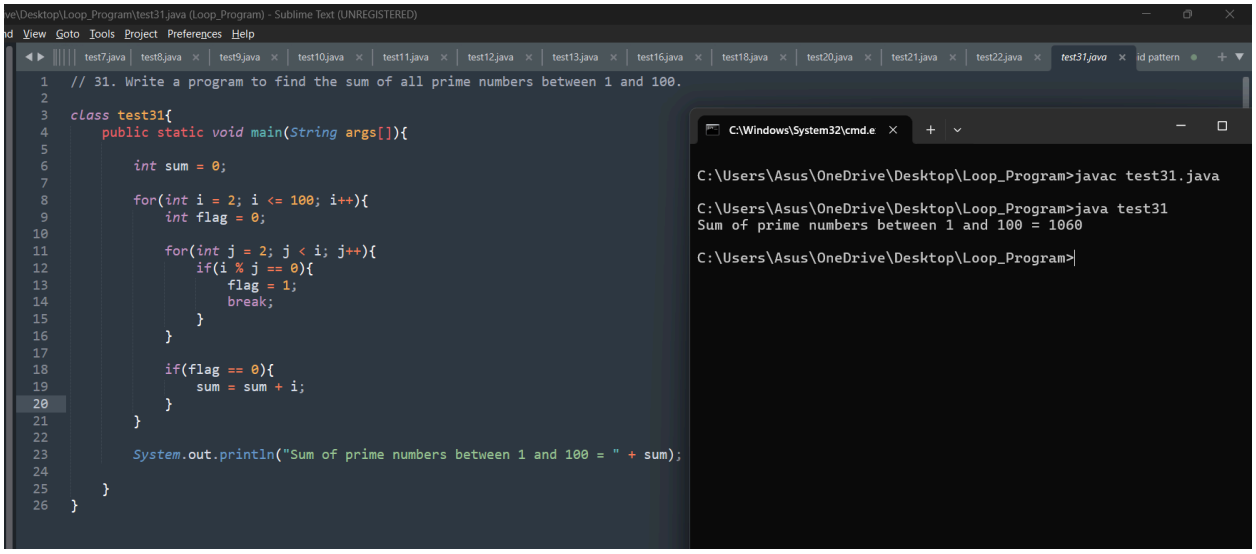
```
class test30{
   public static void main(String args[]){

      int sum = 0;

      for(int i = 1; i <= 10; i++){
         sum = sum + (i * i * i);
      }

      System.out.println("Sum of cubes from 1 to 10 = " + sum);

   }
}
```



// 31. Write a program to find the sum of all prime numbers between 1 and 100.
Solution:

```
class test31{
   public static void main(String args[]){

      int sum = 0;

      for(int i = 2; i <= 100; i++){
         int flag = 0;

         for(int j = 2; j < i; j++){
            if(i % j == 0){
               flag = 1;
               break;
            }
         }

         if(flag == 0){
            sum = sum + i;
```

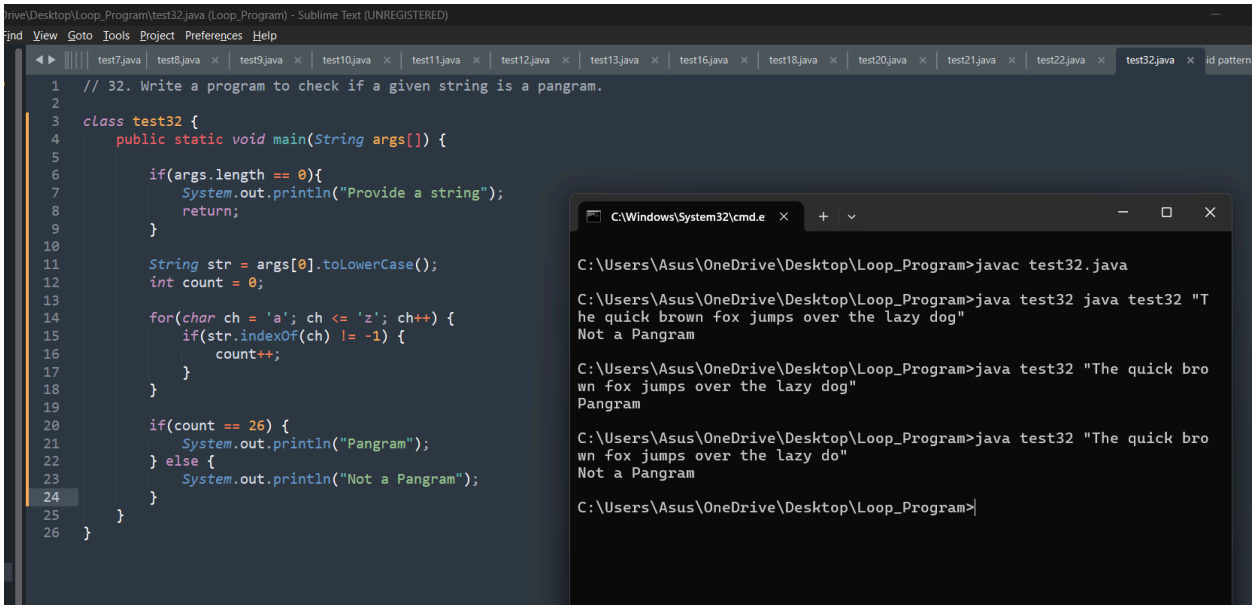
```
        }
    }

    System.out.println("Sum of prime numbers between 1 and 100 = " + sum);

  }
}
```



// 32. Write a program to check if a given string is a pangram.
Solution:

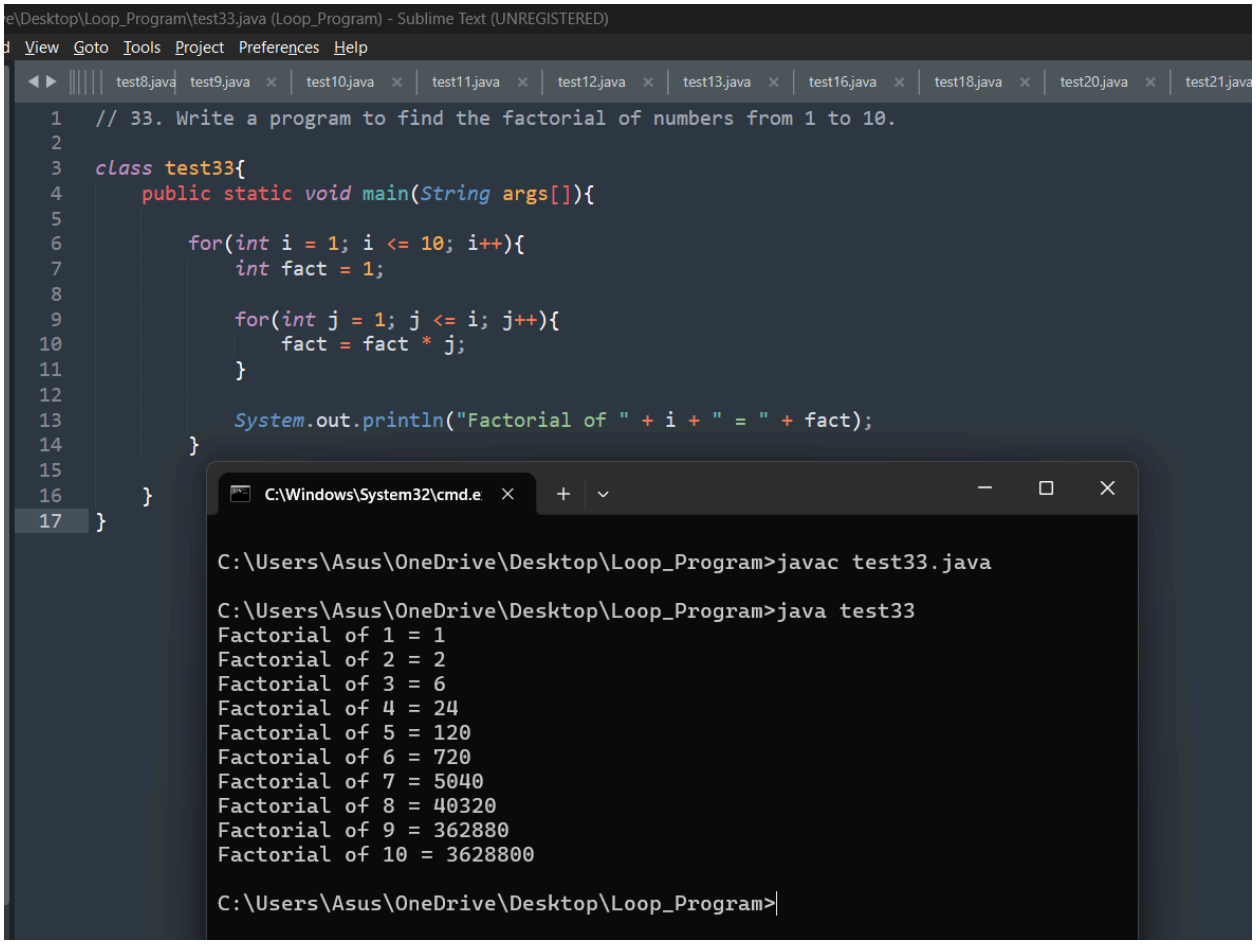
```
class test32{
  public static void main(String args[]){

    String str = args[0].toLowerCase();
    int count = 0;

    for(char ch = 'a'; ch <= 'z'; ch++){
      if(str.indexOf(ch) != -1){
        count++;
      }
    }

    if(count == 26){
      System.out.println("Pangram");
    }else{
      System.out.println("Not a Pangram");
    }
```

```
    }
}
```



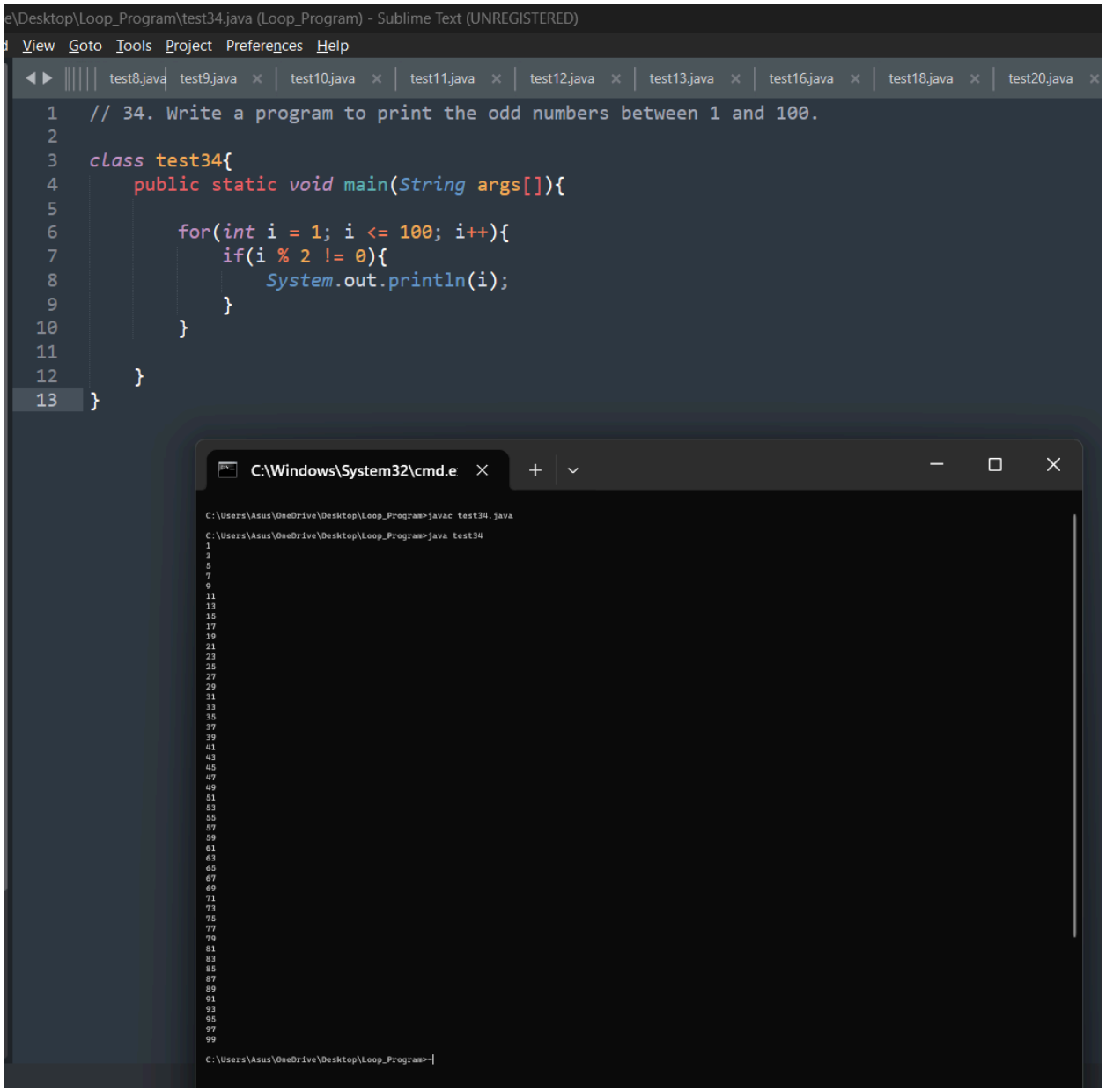
// 33. Write a program to find the factorial of numbers from 1 to 10.
Solution:

```
class test33{
    public static void main(String args[]){

        for(int i = 1; i <= 10; i++){
            int fact = 1;

            for(int j = 1; j <= i; j++){
                fact = fact * j;
            }

            System.out.println("Factorial of " + i + " = " + fact);
        }

    }
}
```



// 34. Write a program to print the odd numbers between 1 and 100.
Solution:

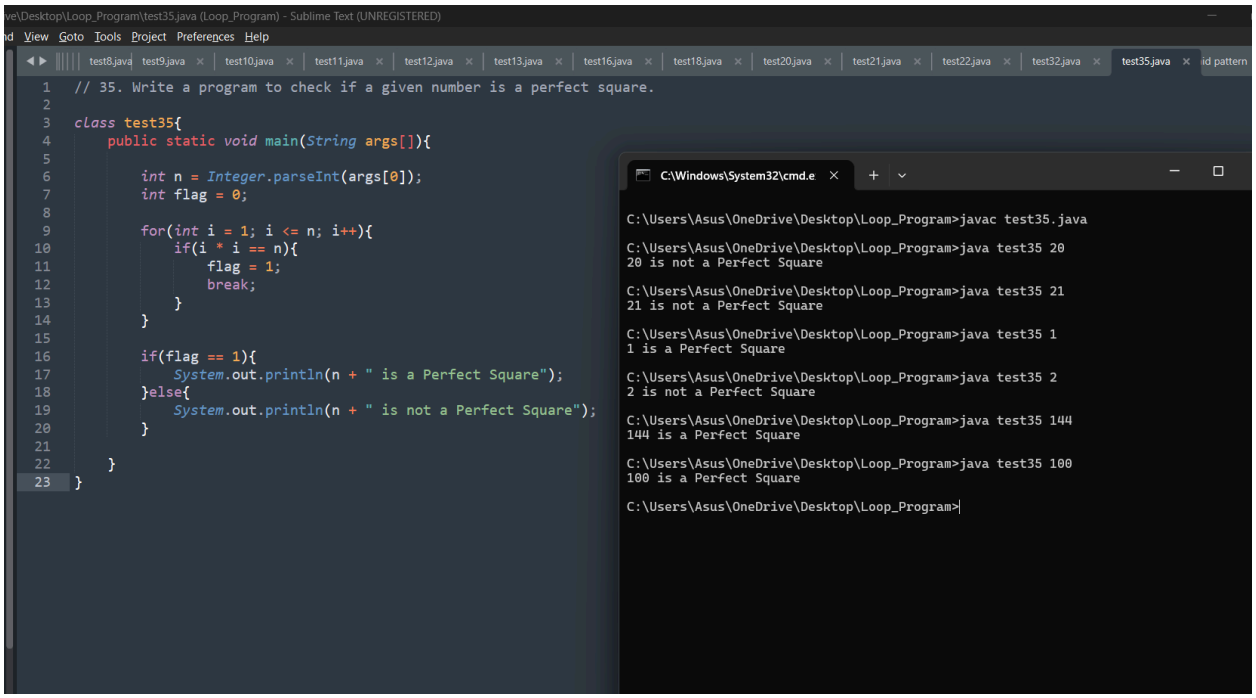
```
class test34{
   public static void main(String args[]){

      for(int i = 1; i <= 100; i++){
         if(i % 2 != 0){
            System.out.println(i);
         }
      }

   }
}
```



// 35. Write a program to check if a given number is a perfect square.
Solution:

```
class test35{
   public static void main(String args[]){

      int n = Integer.parseInt(args[0]);
      int flag = 0;
```

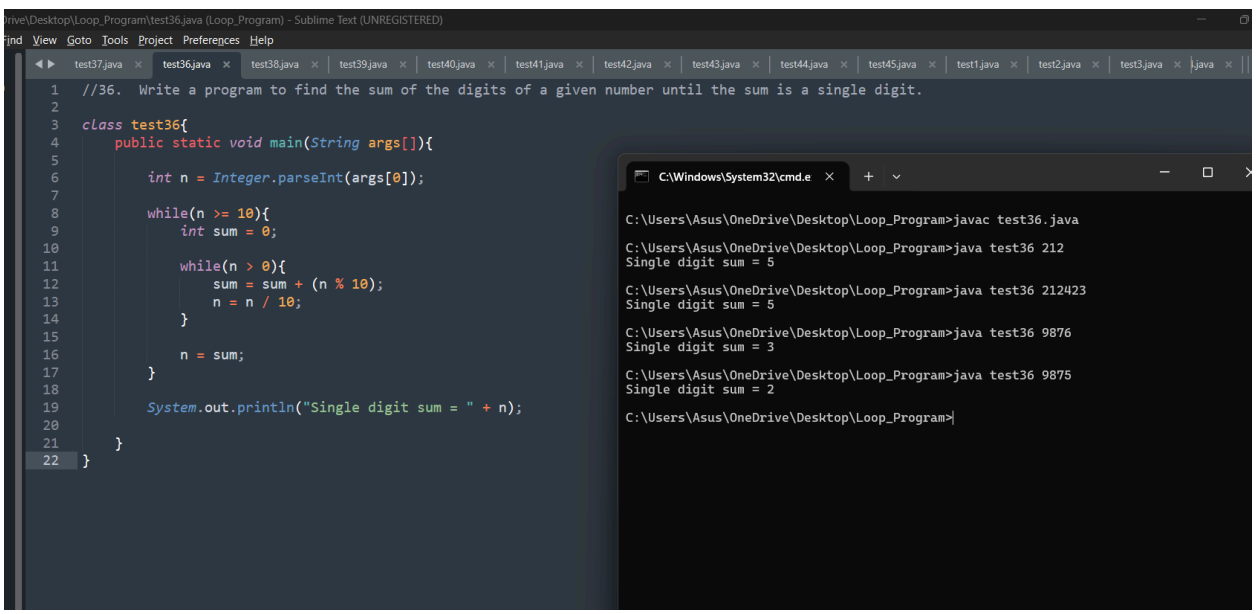
```
        for(int i = 1; i <= n; i++){
            if(i * i == n){
                flag = 1;
                break;
            }
        }

        if(flag == 1){
            System.out.println(n + " is a Perfect Square");
        }else{
            System.out.println(n + " is not a Perfect Square");
        }

    }
}
```



//36. Write a program to find the sum of the digits of a given number until the sum is a single digit.
Solution:

```
class test36{
    public static void main(String args[]){

        int n = Integer.parseInt(args[0]);
```

```
        while(n >= 10){
            int sum = 0;

            while(n > 0){
                sum = sum + (n % 10);
                n = n / 10;
            }

            n = sum;
        }

        System.out.println("Single digit sum = " + n);

    }
}
```

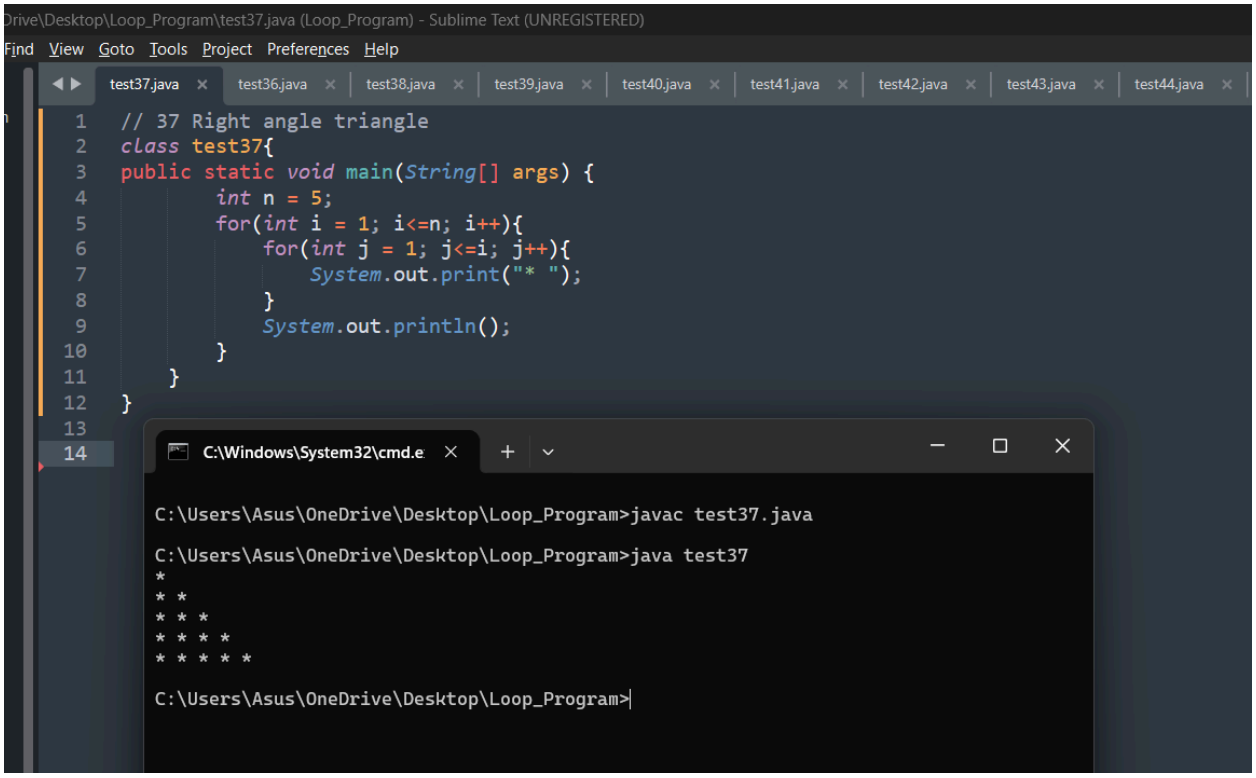


// 37 Right angle triangle

Solution:

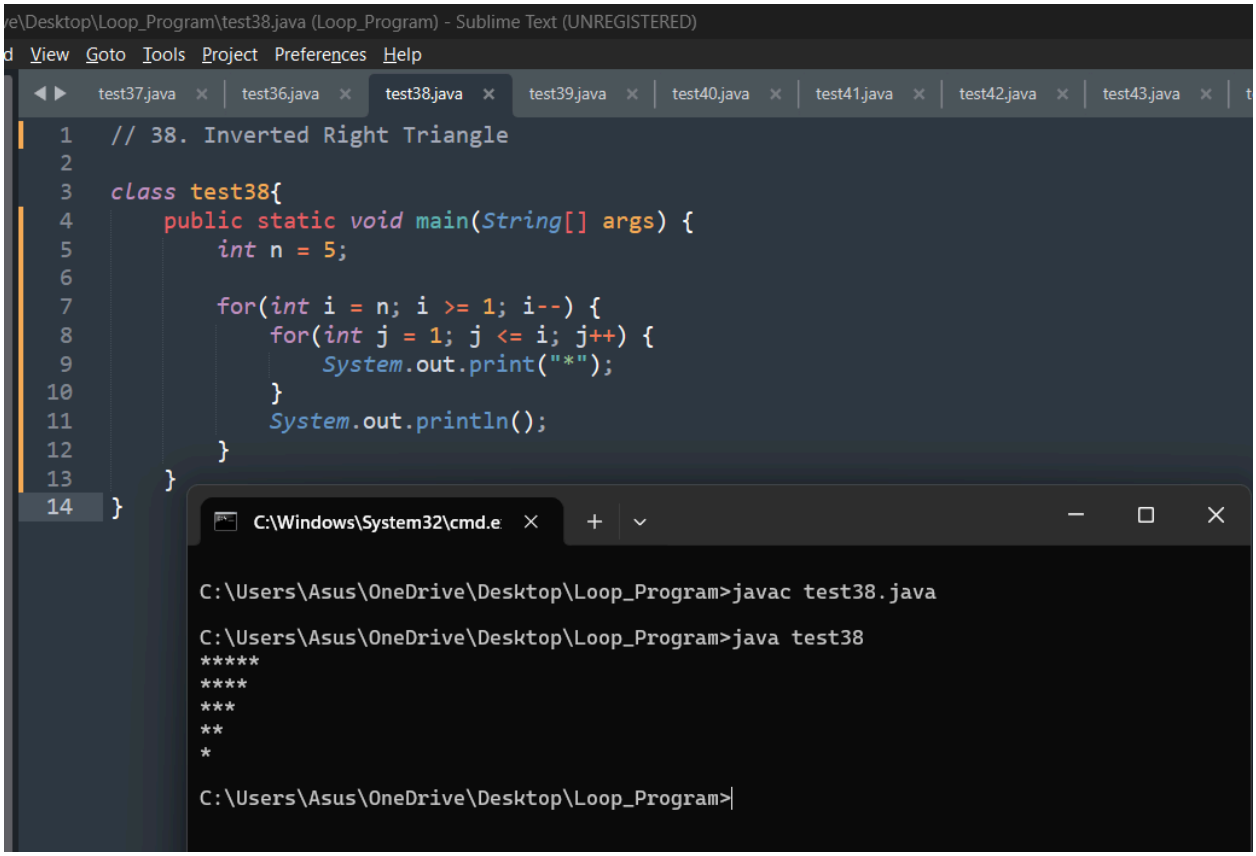
```
class test37{
public static void main(String[] args) {
                int n = 5;
                for(int i = 1; i<=n; i++){
                        for(int j = 1; j<=i; j++){
                                System.out.print("* ");
                        }
```

```
                System.out.println();
            }
        }
}
```



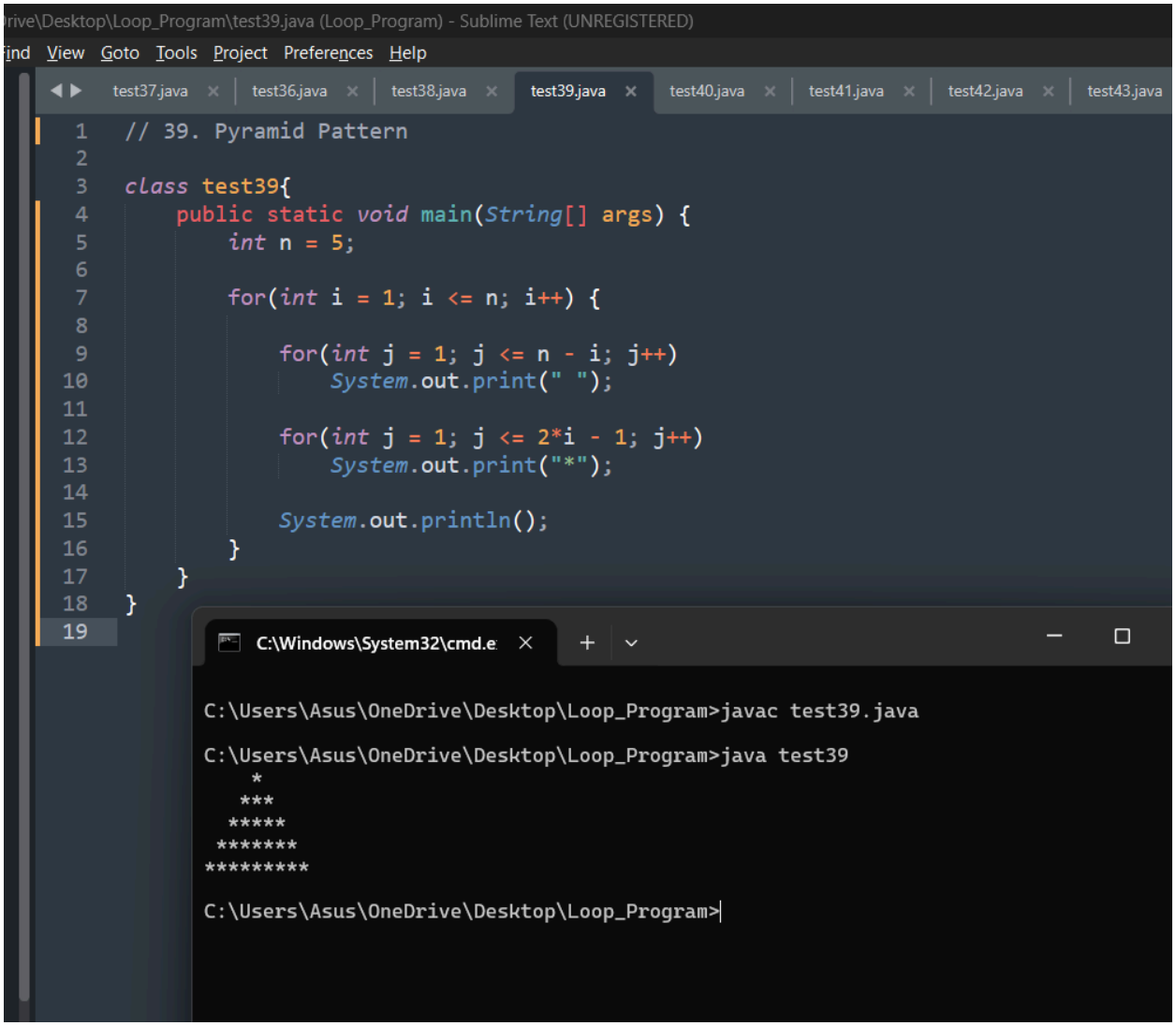
// 38. Inverted Right Triangle
Solution:

```
class test38{
    public static void main(String[] args) {
        int n = 5;

        for(int i = n; i >= 1; i--) {
            for(int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```



// 39. Pyramid Pattern
Solution:

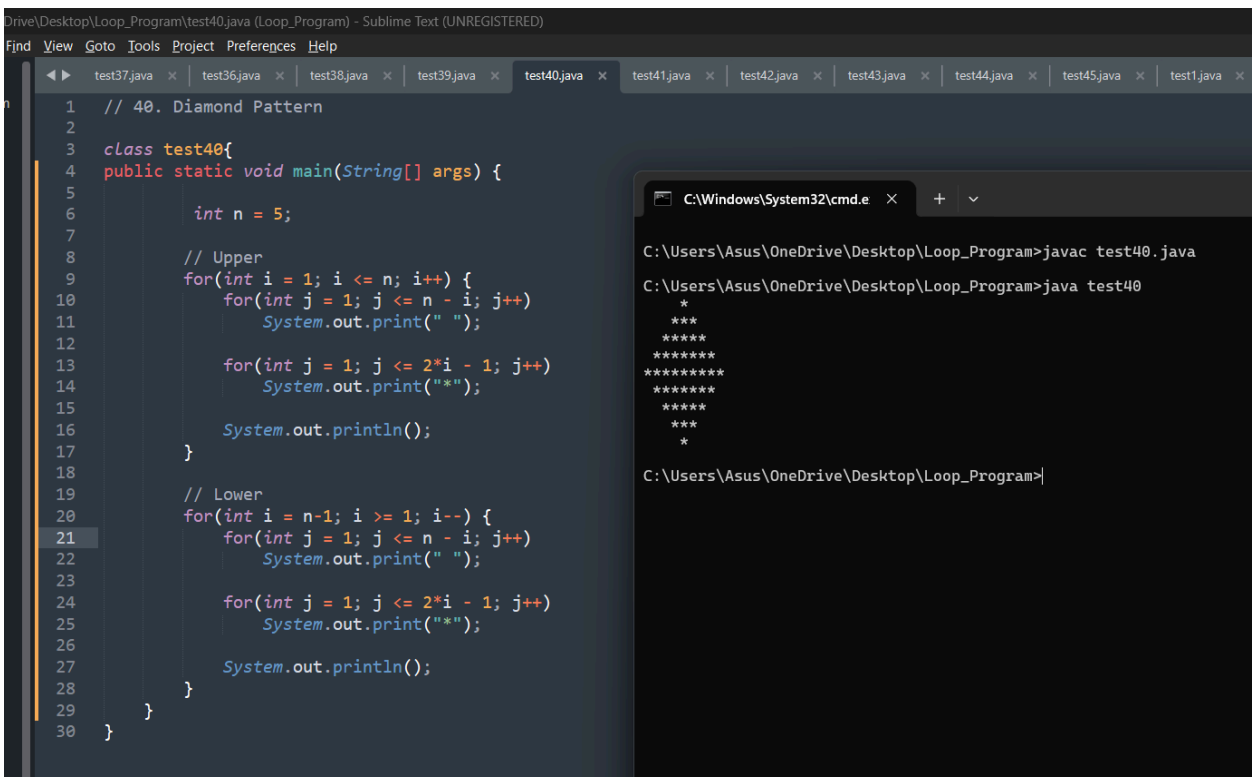
```
class test39{
        public static void main(String[] args) {
      int n = 5;

      for(int i = 1; i <= n; i++) {

         for(int j = 1; j <= n - i; j++)
            System.out.print(" ");

         for(int j = 1; j <= 2*i - 1; j++)
            System.out.print("*");

         System.out.println();
      }
   }
}
```



// 40. Diamond Pattern
Solution:

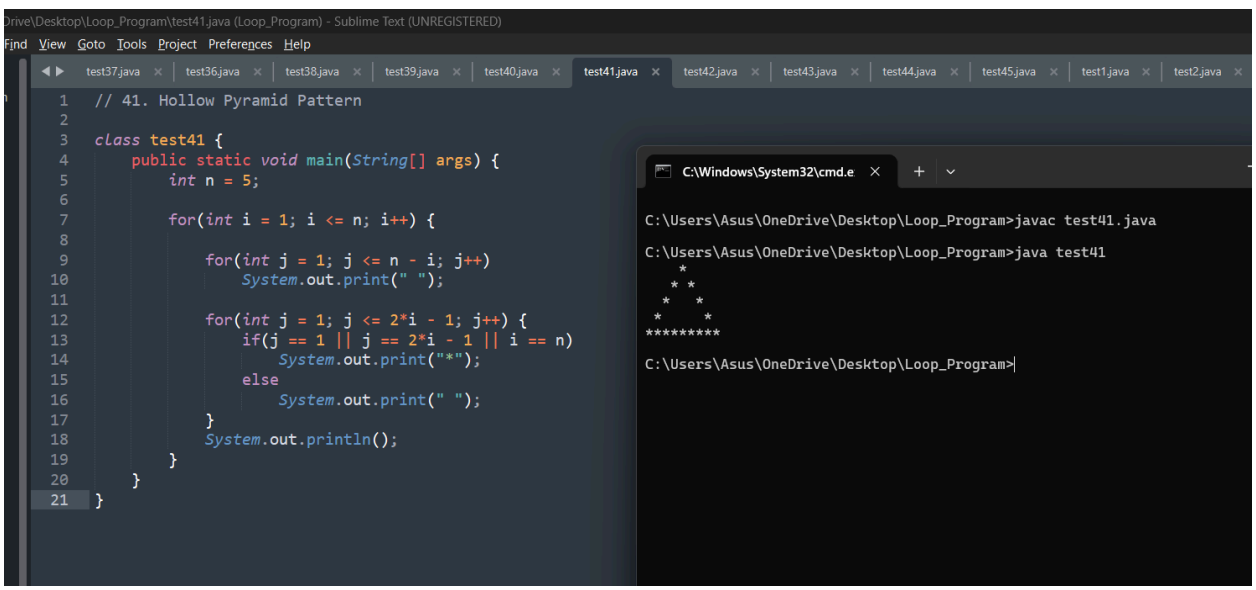
```
class test40{
public static void main(String[] args) {

        int n = 5;

    // Upper
    for(int i = 1; i <= n; i++) {
        for(int j = 1; j <= n - i; j++)
            System.out.print(" ");

        for(int j = 1; j <= 2*i - 1; j++)
```

```
            System.out.print("*");

        System.out.println();
    }

    // Lower
    for(int i = n-1; i >= 1; i--) {
        for(int j = 1; j <= n - i; j++)
            System.out.print(" ");

        for(int j = 1; j <= 2*i - 1; j++)
            System.out.print("*");

        System.out.println();
    }
  }
}
```



// 41. Hollow Pyramid Pattern
Solution:

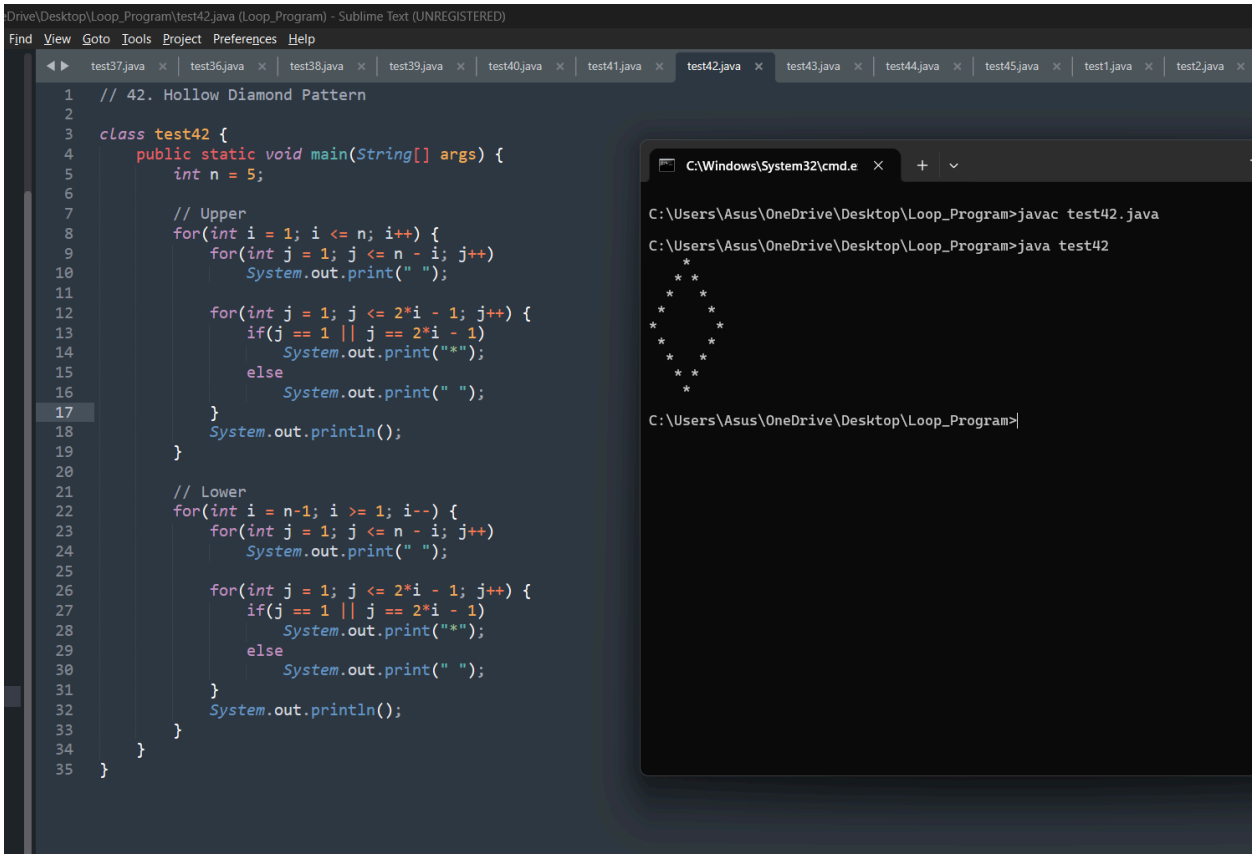
```
class test41 {
    public static void main(String[] args) {
        int n = 5;

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= n - i; j++)
                System.out.print(" ");

            for(int j = 1; j <= 2*i - 1; j++) {
                if(j == 1 || j == 2*i - 1 || i == n)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



// 42. Hollow Diamond Pattern
Solution:

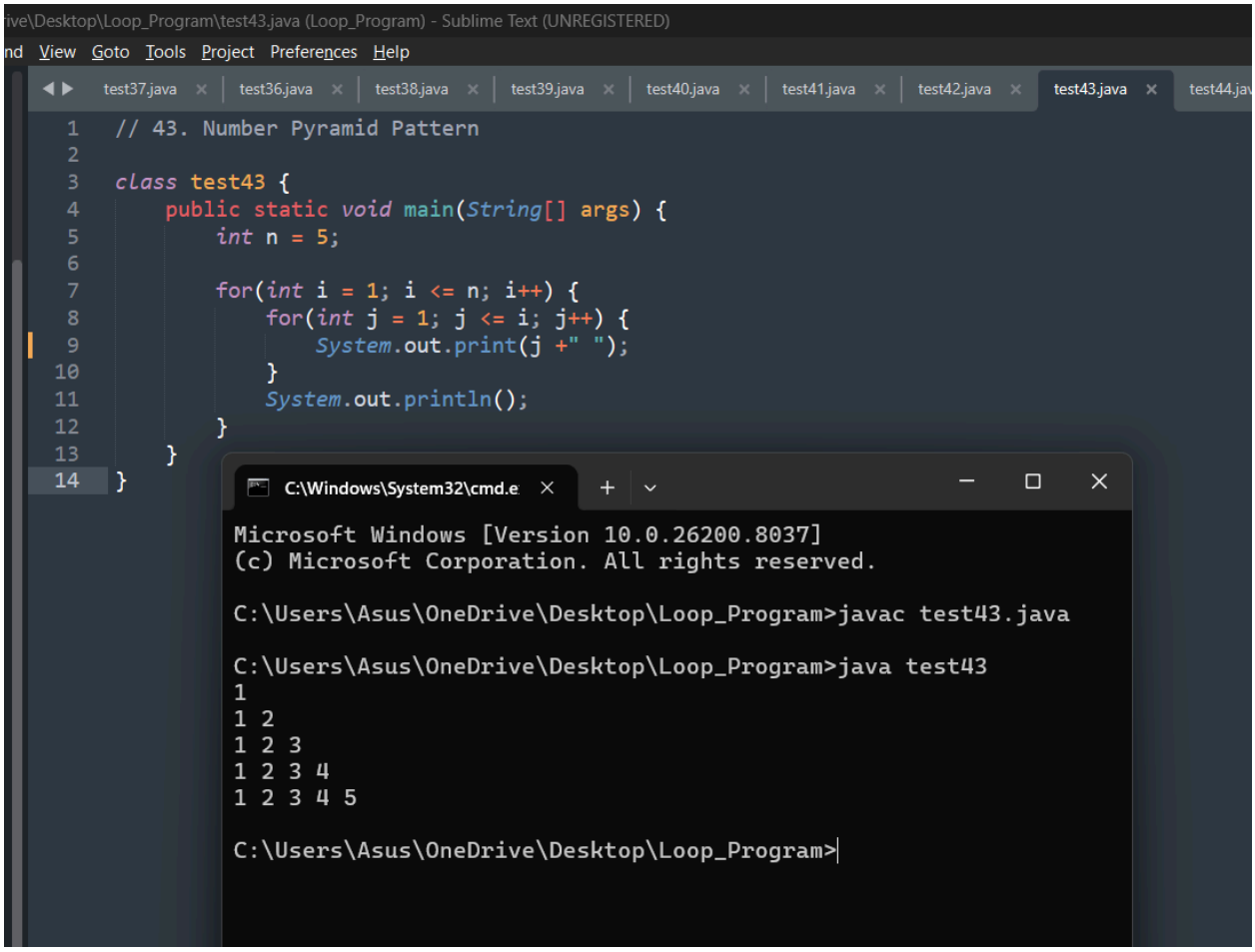
```
class test42 {
    public static void main(String[] args) {
        int n = 5;
```

```
        // Upper
        for(int i = 1; i <= n; i++) {
            for(int j = 1; j <= n - i; j++)
                System.out.print(" ");

            for(int j = 1; j <= 2*i - 1; j++) {
                if(j == 1 || j == 2*i - 1)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }

        // Lower
        for(int i = n-1; i >= 1; i--) {
            for(int j = 1; j <= n - i; j++)
                System.out.print(" ");

            for(int j = 1; j <= 2*i - 1; j++) {
                if(j == 1 || j == 2*i - 1)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



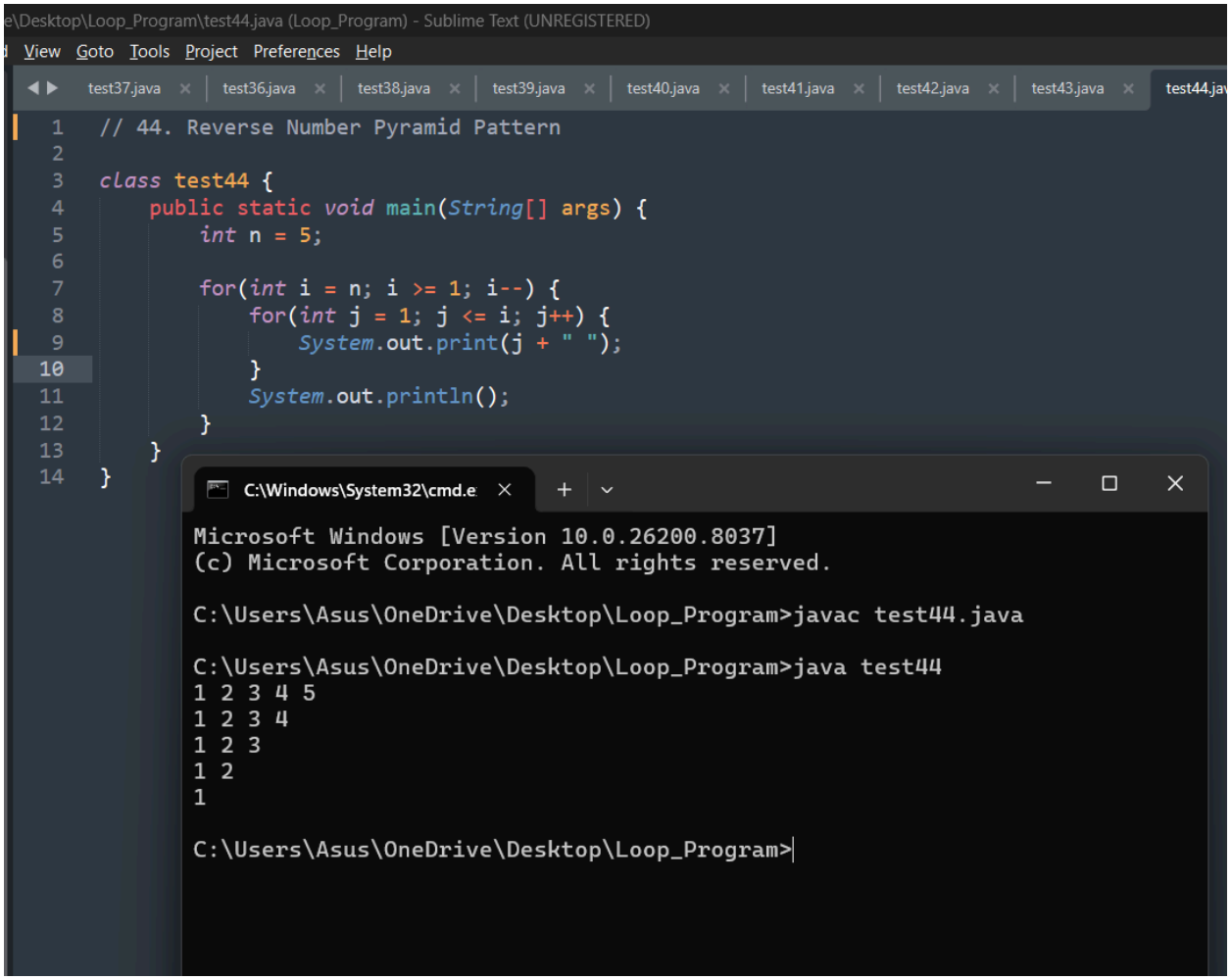
// 43. Number Pyramid Pattern
Solution:

```
class test43 {
   public static void main(String[] args) {
      int n = 5;

      for(int i = 1; i <= n; i++) {
         for(int j = 1; j <= i; j++) {
            System.out.print(j);
         }
         System.out.println();
      }
   }
}
```



// 44. Reverse Number Pyramid Pattern
Solution:

```
class test44 {
   public static void main(String[] args) {
      int n = 5;

      for(int i = n; i >= 1; i--) {
         for(int j = 1; j <= i; j++) {
            System.out.print(j);
         }
         System.out.println();
      }
   }
}
```



// 45. Floyd's Triangle Pattern
Solution:

```
class test45{
   public static void main(String[] args) {
      int n = 4;
      int num = 1;

      for(int i = 1; i <= n; i++) {
         for(int j = 1; j <= i; j++) {
            System.out.print(num + " ");
            num++;
         }
         System.out.println();
      }
   }
}
```

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test36.ja | test38.java | test39.java | test40.java | test41.java | test42.java | test43.java | test44.java | test45.ja

```java
// 45. Floyd's Triangle Pattern

class test45{
    public static void main(String[] args) {
        int n = 4;
        int num = 1;

        for(int i = 1; i <= n; i++) {
            for(int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }
            System.out.println();
        }
    }
}
```

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```
C:\Users\Asus\OneDrive\Desktop\Loop_Program>javac test45.java

C:\Users\Asus\OneDrive\Desktop\Loop_Program>java test45
1
2 3
4 5 6
7 8 9 10

C:\Users\Asus\OneDrive\Desktop\Loop_Program>
```